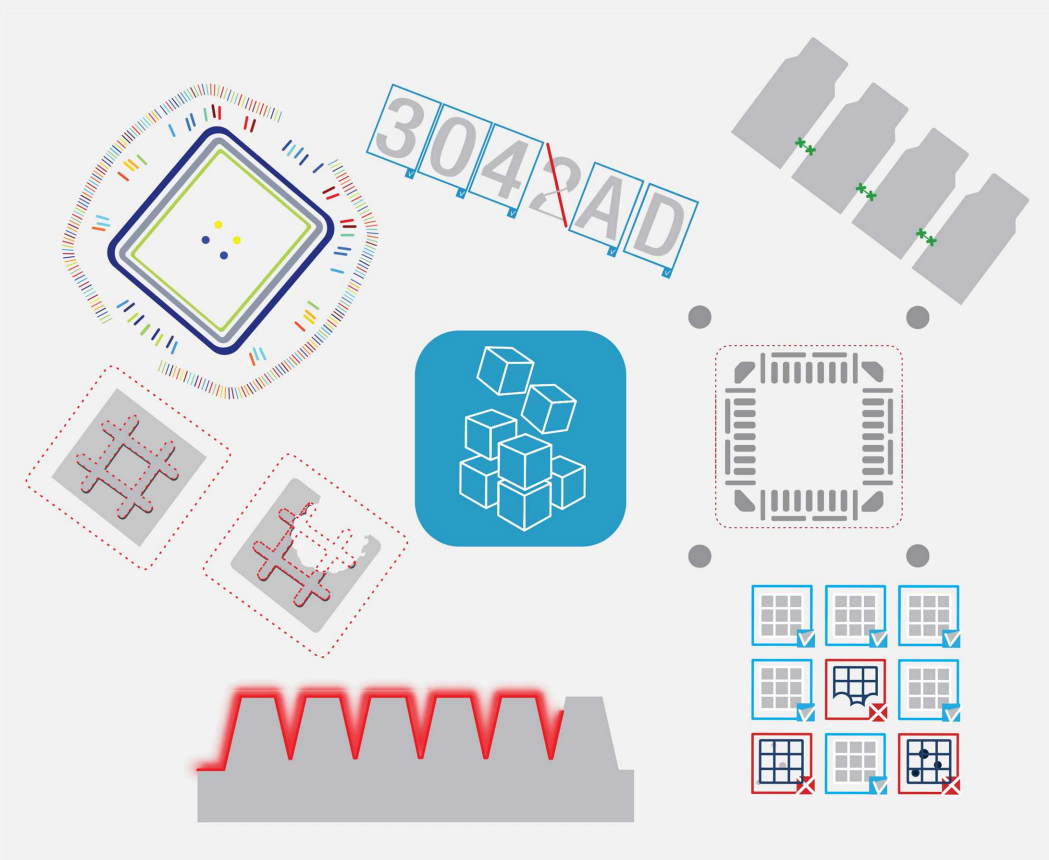




사용자 설명서

Open eVision Studio



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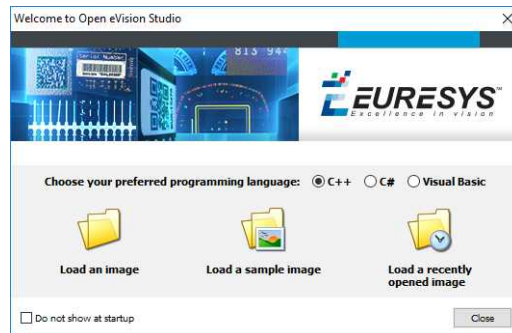
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1. Open eVision Studio 사용하기

1.1. 프로그래밍 언어를 선택하십시오.

오픈 시작할 때 Open eVision Studio 다음과 같은 웰 스크린을 표시합니다.



1. 프로그래밍 언어를 선택하십시오.

✓ 선택한 것을 저장하고 다음 시작 시에는 사용자의 프로그래밍 언어를 자동으로 선택합니다 Open eVision Studio.

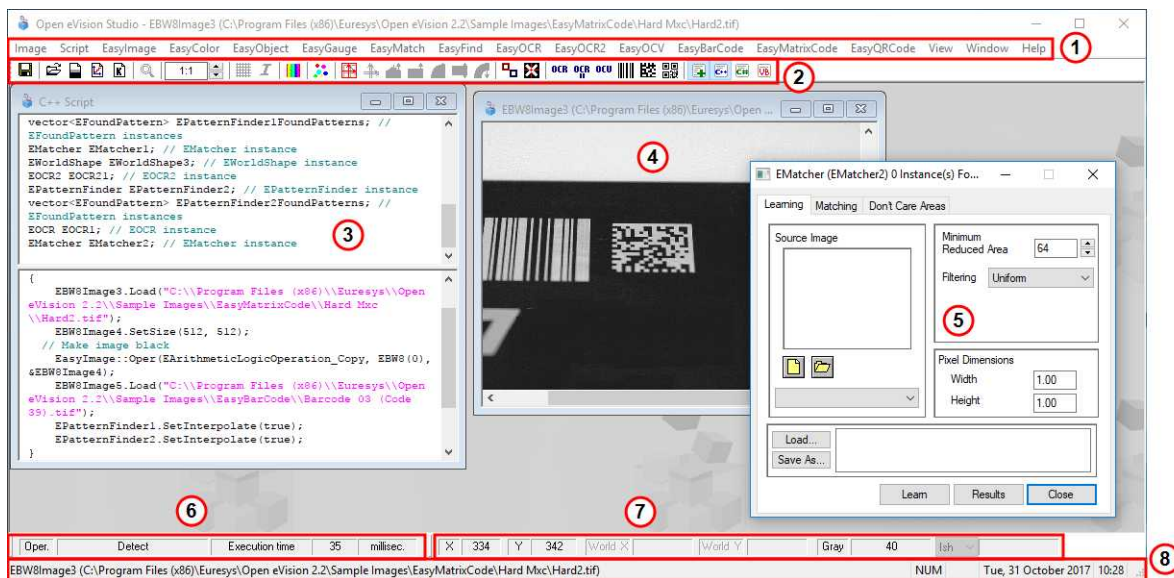
📁 프로그래밍 언어를 바꾸면 스크립트 윈도우에 있는 어떤 스크립트든지 자동으로 삭제되고 윈도우 내용이 기화됩니다.

2. 이미 불러들인 한 개 혹은 몇 개의 이미지를 나중에 리하기 위해서 불러오기 버튼 중 하나를 클릭하십시오.

3. 다음 시작시 웰 스크린을 보고 싶지 않으면 시작할 때 보지 않기 박스를 하십시오. Open eVision Studio.

✓ 웰 스크린을 보려면 언제든지 **도움** > **웰컴 스크린** 메뉴로 가서 설정을 변경하십시오.

1.2. 인터페이스 둘러보기



Open eVision Studio 그래픽 유저 인터페이스 (GUI) 는 다음과 같이 이루어져 있습니다:

1. **메인 메뉴 바**에서 모든 라이브러리의 기능과 툴을 이용할 수 있습니다.

☑ Open eVision Studio 라이선스가 필요하지 않으며 모든 라이브러리를 시험해볼 수 있습니다. 물론 Open eVision Studio 에서 사용자의 애플리케이션에 드를 복사하였지만 필요한 라이선스가 없으면 실행시 “라이선스 없음” 오류 메시지를 보여줍니다.

2. **주요 도구 모음**을 사용하면 이미지, 모양, 게이지, 바코드, 매트릭스 등과 같은 주요 Open eVision 개체에 빠르게 액세스할 수 있습니다.
3. **스립트 윈도우**는 Open eVision Studio에서 수행한 작업에 해당하는 드를 선택한 프로그래밍 언어로 표시합니다. 언제든지 이 드를 자신의 응용 프로그램에 저장하거나 복사할 수 있습니다.
4. **이미지 윈도우**에는 라이브러리 및 도구를 사용하여 리할 수 있는 오픈 이미지가 표시됩니다.
5. **도구 윈도우**를 사용하여 사용 가능한 모든 도구를 쉽게 구성할 수 있습니다. 해당 설정이 스텑 윈도우에 자동으로 가되어 쉽게 재사용할 수 있습니다.

☑ 대부분의 도구 은 유동적이어서 화면 기를 보다 잘 활용하기 위해 Open eVision Studio 기본 밖으로 쉽게 이동할 수 있습니다.

6. **실행 시간 막대**는 퓨에서 선택한 기능 (밀리 또는 마이크로 단위로 정) 실행에 소요된 정확한 시간을 표시합니다. 이 정확한 정은 애플리케이션의 성능을 평가하는 데에 도움이 됩니다.
7. **색상 도구 모음**에는 이미지 서의 X 및 Y 좌표와 해당 픽셀 값과 같은 현재 정보가 표시됩니다.
8. **상 표시 줄**에는 활성 이미지 파일 경로와 같은 애플리케이션에 대한 일반 정보가 표시됩니다.

1.3. 이미지에서 도구를 실행하기

1단계 : 도구 구성하기

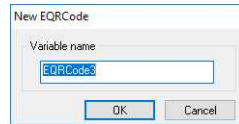
Open eVision Studio를 사용할 때 번째 단계는 이미지에 사용할 라이브러리와 도구를 선택합니다

그렇게 하려면 :

1. 기본 메뉴 표시 줄에서 사용할 라이브러리를 클릭하십시오.
2. 사용할 도구를 클릭하십시오.

모든 라이브러리 (EasyImage, EasyColor 및 EasyGauge 제외)는 New Xxx Tool이라는 하나의 도구만 표시합니다. 이 라이브러리 중 일부는 추가 기능도 제공합니다.

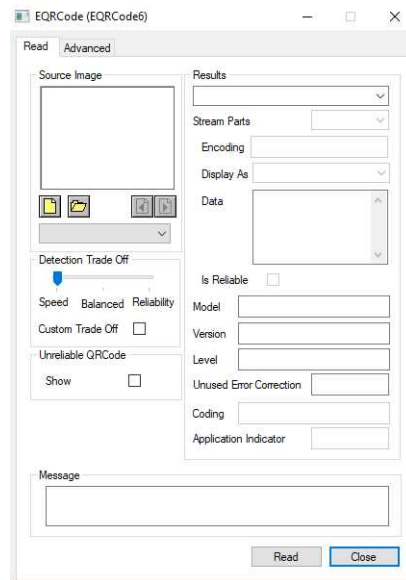
3. 대화 상자에서 자동으로 생성되고 결과에 포함될 변수의 Variable name 을 입력하십시오.



EasyQRCode에 대한 변수 생성 대화 상자의 예

4. Ok를 클릭하십시오.

선택한 도구 대화 상자가 열립니다.



EasyQRCode에 대한 변수 생성 대화 상자의 예

다음 단계는 "2단계 : 이미지 열기"페이지6.

2단계 : 이미지 열기

라이브러리와 도구를 선택했다면 이 도구를 적용할 이미지를 열어야 합니다.

선택한 도구 대화 상자의 소스 이미지 영역에서 다음을 수행합니다.

1. 이미지 열기:

- Click on the  이미지 열기 버튼을 클릭하고 파일에서 하나 이상의 이미지 (SHIFT 및 CTRL 사용)를 선택하십시오.
- 또는 이미 드롭 다운 목록에서 열려있는 이미지 중 하나 (또는 ROI가 있는 경우) 중 하나를 선택하십시오.

적절한 파일 형식 (JPG, PNG, TIFF 또는 BMP) 및 라이브러리에 따라 8비트 및 / 또는 24비트의 이미지만 선택할 수 있습니다.



2. 이미지를 여러 개 선택한 경우 Load Previous 또는 Load Next 버튼으로 이미지를 활성화하십시오.

이 도구는 로드된 이미지에 자동으로 적용되며, 이 단계에서 도구 기본 설정에 따라 결과가 표시됩니다.

다음 단계는 "3단계 : ROI 관리하기" 페이지 7.

3단계 : ROI 관리하기

몇 가지 경우에, 대부분 처리 시간을 줄이거나 읽으려는 대상을 단일화하기 위해서는, 전 이미지를 처리하는 것이 아니라 이미지의 하나 또는 여러 개의 잘 정의된 사각형 부분 또는 ROI (관심 영역)를 처리하기를 원하지 않을 것입니다.

✓ Open eVision에서 ROI는 이미지에 부되어 있으며 부모 이미지가 있는 동안에만 존재합니다.

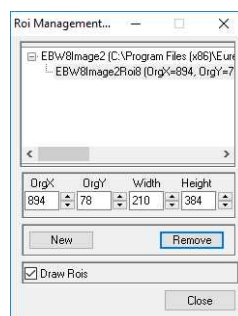
ROI 생성하기

1. 이미지 열기:

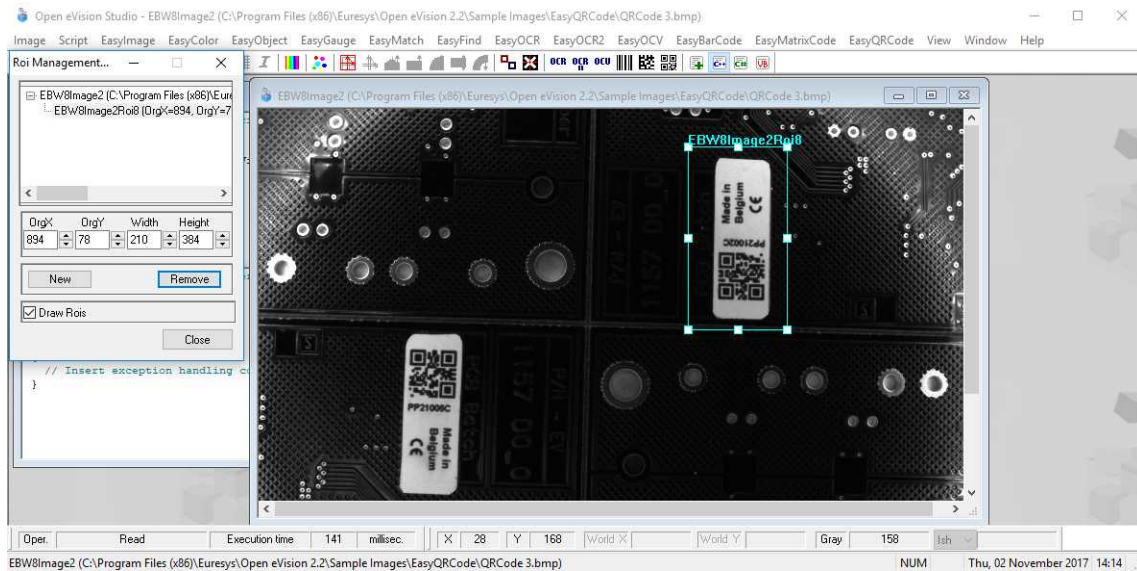
- 이미지가 이미 열려 있으면 해당 이미지를 활성화하십시오.
- 이미지가 아직 열리지 않은 경우 기본 메뉴로 이동하십시오: 열기 위해서 **Image** > **Open...**

2. ROI를 만들려면 기본 메뉴로 이동하십시오: **Image** > **ROI Management...**

ROI Management 윈도우는 아래 보이는 대로 표시됩니다.



3. 트리에서 이미지를 선택하십시오.
4. **New** 버튼을 클릭하십시오.
5. 대화 상자에서 새 RIO를 위한 **Variable name**를 넣으십시오.
- ROI는 아래 그림과 같이 이미지에 색상 사각형으로 표시됩니다.



6. ROI 모서리와 면 핸들을 끌어 필요한 위치로 이동하십시오.

7. ROI Management 윈도우를 닫기 위해 Close 버튼을 클릭하십시오.

다음 단계는 "4 단계 : 도구 구성하기"페이지9.

ROI 관리하기

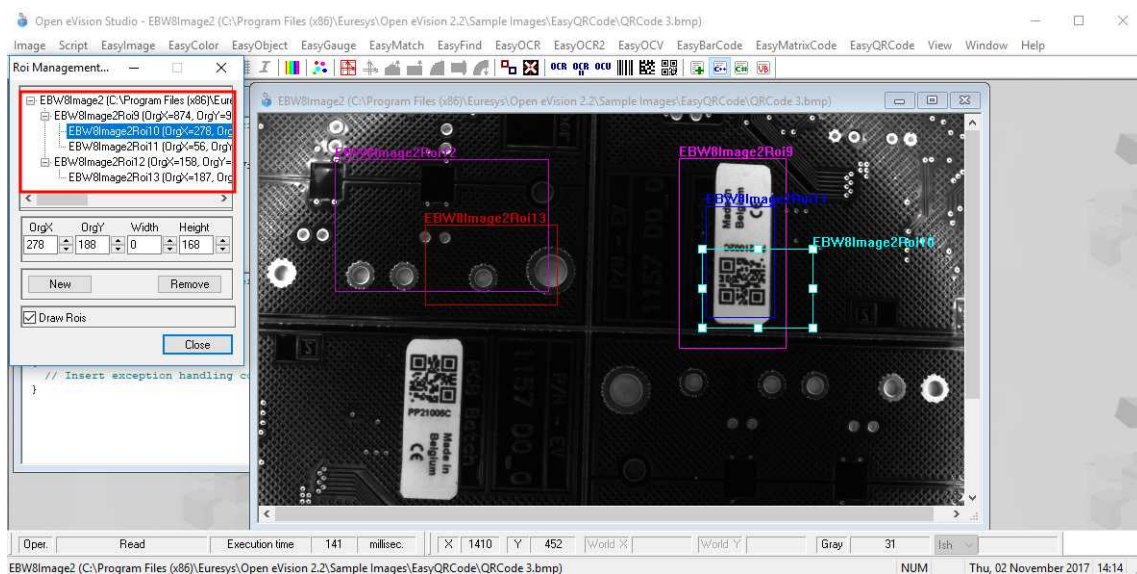
ROI를 추가, 변경 및 제거할 수 있습니다.

☑ 이미지는 여러 개의 ROI를 가질 수 있습니다. 각 ROI는 이미지에 직접 부칠 수 있습니다 (즉, 위 ROI가 이미지와 관련됨) 또는 다른 ROI (이 ROI가 '부모' ROI와 관련이 있음을 의미함)에 부칠 수 있습니다.

1. ROI를 관리하려면, 기본 메뉴로 이동하십시오: Image > ROI Management...

ROI Management 윈도우가 아래와 같이 ROI 관계 트리와 함께 표시됩니다.

Draw Rois 상자가 선택되면 모든 ROI가 이미지에 다른 색상으로 표시됩니다.



2. ROI 관계 트리에서 ROI를 선택하십시오.

- ROI 모서리와 면 핸들을 드래그하여 선 한 ROI의 위와 기 (해당하는 경우 모든 ROI의 위)를 변경하십시오.
- New** 클릭하면 선 한 ROI에 부된 새로운 ROI를 가 할 수 있습니다.
 ✓ ROI를 이미지에 직접 부하려면 ROI 관계 트리의 상단에서 이미지를 선 하십시오.
- 선 한 ROI (있는 경우 모든 해당 ROI가 부됩니다)를 삭제하려면 **Remove** 버튼을 클릭하십시오.
- ROI Management** 윈도우를 닫기 위해 **Close** 버튼을 클릭하십시오.

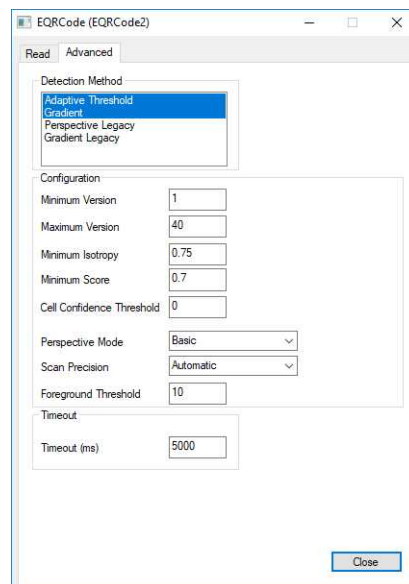
4 단계 : 도구 구성하기

일단 ROI를 작성했다면 ROI를 포함하여 이미지가 준비되면 도구를 구성해야 합니다.

도구 윈도우에서:

- 여러 을여십시오.

✓ 새 도구를 생성하면 모든 매개 변수가 기본값으로 설정됩니다.



EasyQRCode 도구의 매개 변수 예제

- 각 에서 원하는대로 매개 변수의 값을 설정하십시오.

매개 변수, 기능 및 기본값에 대한 자세한 내용은 "기능 안내서" 및 "레퍼런스 매뉴얼"을 조하십시오.

학습 또는 게이지 사용과 같은 특정 작업에 대해서는 "기능 안내서"를 조하십시오.

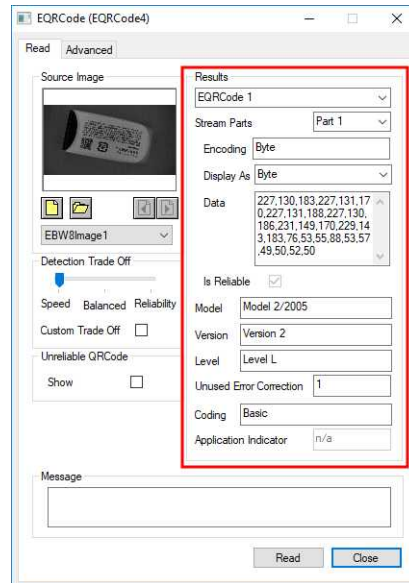
- 다음 단계에서 설명한대로 도구를 실행하고 결과를 분석하십시오 ["5 단계 : 도구 실행 및 실행 시간 확인하기" 페이지 9](#).

5 단계 : 도구 실행 및 실행 시간 확인하기

도구 매개 변수가 설정되면 도구를 실행하고 원하는 경우 퓨 에서 실행 시간을 확인하십시오.

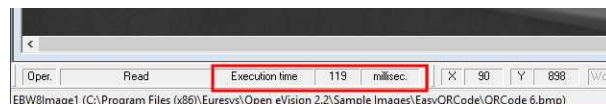
도구 윈도우에서:

1. 선택한 이미지에서 도구를 실행하려면 **Read**, **Detect**, **Results** 혹은 **Execute** 버튼을 클릭하십시오.(라이브러리 기능에 따라 다름)
2. 아래 그림과 같이 이미지와 결과 필드 또는 영역에서 결과를 확인하십시오.



QRCode를 읽은 결과의 예

3. 예상되는 결과가 없는 경우:
 - 매개 변수를 변경하십시오 (기본값으로 시작한 다음 한 번에 하나의 매개 변수만 변경하십시오).
 - 이미지가 분해되지 않은 경우, 예 설정된대로 이미지를 강화하십시오.
4. 기본 Open eVision Studio 윈도우의 왼쪽 하단에 있는 실행 시간 막대에서 실행 시간을 확인하십시오.



실행 시간

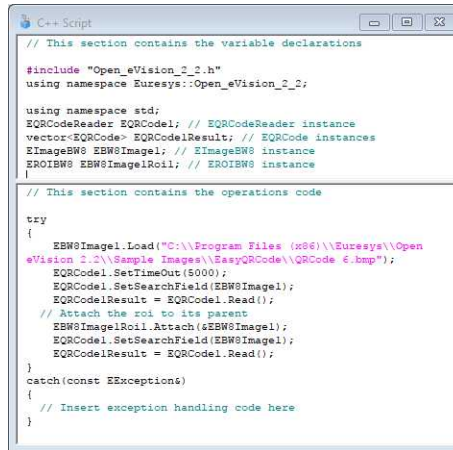
☑ 실행 시간은 퓨에서 정한 실제 리 시간입니다. 그것은 퓨 프로세서, 메모리, 운영 체제 및 실행시 프로세서 로드 등에 따라 다릅니다. 따라서 이 실행 시간은 실행마다 약간씩 다릅니다.

5. 보다 대표적인 실행 시간을 얻으려면 **Read**, **Detect**, **Results** 혹은 **Execute** 여러 번 클릭하고 평균 실행 시간을 계산하십시오.
6. 애플리케이션에서 실행 시간을 줄여야 하는 경우 다음을 시도하십시오.
 - 도구 매개 변수를 변경하려면,
 - 이미지에 하나 이상의 ROI를 추가하려면,
 - 이미지 향상하기 위해

다음 단계는 "6단계: 생성된 코드 사용하기" 페이지 10.

6단계: 생성된 코드 사용하기

기본적으로 Open eVision Studio 인터페이스에서 수행하는 모든 작업을 아래 그림과 같이 선택한 언어의 드로 변환합니다.



도구 결과가 적합하면 이 생성된 드를 저장하거나 복사하여 자신의 애플리 이션에서 사용할 수 있습니다.

애플리 이션에 드 복사 및 붙여 넣기

스 립트 윈도우에서:

1. 복사할 드 섹션을 선택 하십시오.
2. 이 드에서 오른쪽 클릭하고 메뉴에서 **복사**를 클릭하십시오.
3. 개발 환경 도구로 가서 드를 제자리에 붙여 넣으십시오.

드를 저장하십시오.

1. 스 립트 메뉴로 가십시오.
2. 다른 스크립트로 저장...을 클릭하십시오.
3. 드를 스트 파일로 저장하려면 파일 이름과 경로를 입력하십시오.

생성된 드 관리하기

스 립트 메뉴에서:

- 프로그래밍 언어를 선택 하십시오 (언어를 변경하면 스 립트 가 자동으로 삭제됩니다).
- 스크립트 코드 생성을 활성화하거나 비활성화하십시오. 드로 저장하지 않고 일부 작업을 수행하려면 이 옵션을 비활성화하십시오.

1.4. 이미지 전처리 및 저장

언제 이미지를 사전 리해야합니까?

물론, 적의 상황은 이미지를 쉽고 잘 리 할 수 있도록 이미지 수집 시스 을 설정하여 Open eVision 도구를 원활하고 효율적으로 실행하는 것입니다.

이것이 가능하지 않거나 쉽게 할 수없는 경우 실행하고 싶은 Open eVision 도구를 위해서 이미지 또는 ROI를 사전 리하여 강조하고 대비해 둘 수 있습니다.

다양한 사용 가능한 기능을 사용하여 이미지의 이득과 오프셋을 조정하고, 볼루션을 적용하고, 임계값을 조정하고, 이미지를 회전 및 화이트 밸런스 조정하고, 윤곽선을 강조할 수 있습니다... EasyImage 와 EasyColor 기능을 사용합니다.

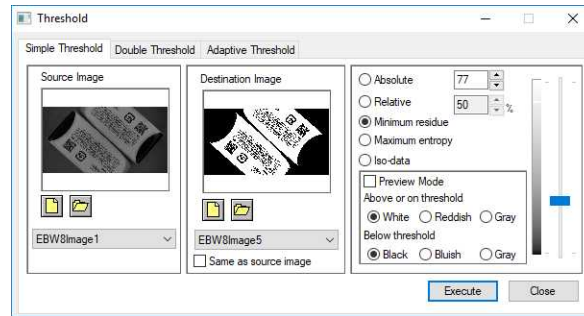
이미지 전 리

이미지 전 리와 실행중인 도구의 이점은 전 리가 새로운 이미지를 생성하는 반면 도구는 주로 이미지를 변경하지 않고 이미지에서 및 검색한다는 점입니다.

이미지나 ROI 전 리하기:

1. 기본 메뉴 표시 줄에서 사용하려는 라이브러리 (EasyImage 또는 EasyColor)를 클릭하십시오.
2. 사용할 기능을 클릭하십시오.

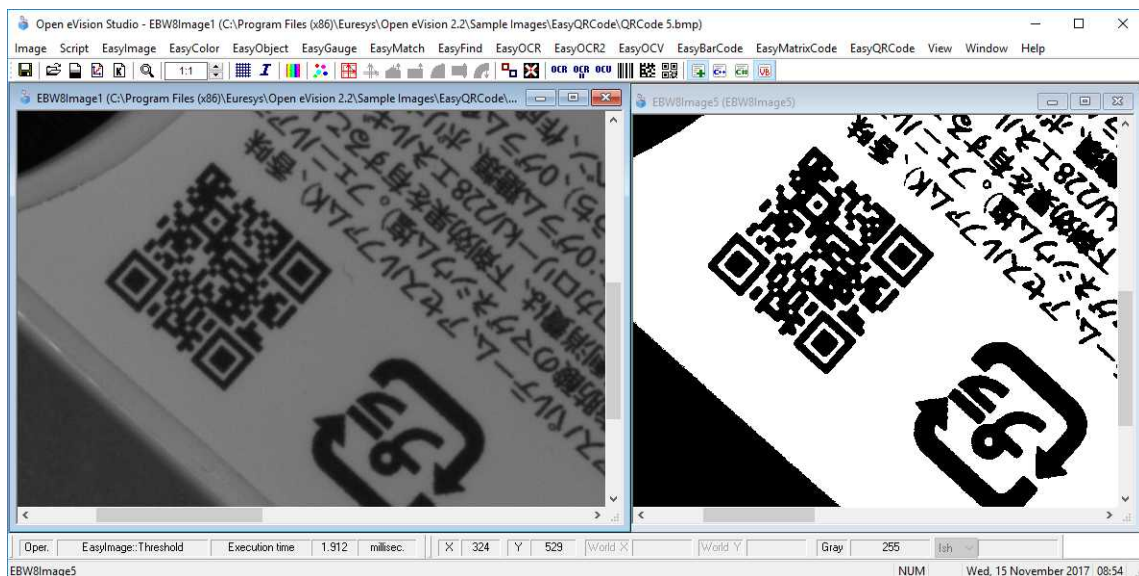
대부분의 기능 대화 상자는 아래 그림과 같이 2 개의 이미지 선택 영역과 매개 변수 설정 영역으로 비슷하게 되어 있습니다.



사전 리 대화 상자의 예 (EasyImage와 임계값)

3. 선택한 기능에 여러 버전이 있는 경우 해당 버전을 엽니다.
4. **소스 이미지** 영역에서, 소스 이미지를 엽니다. ("2단계 : 이미지 열기"페이지6에서 설명한 대로).
5. **대상 이미지** 영역에서, 새로운 대상 이미지를 열거나 만드십시오.
6. 매개 변수를 설정하십시오.
7. **실행** 버튼을 클릭하십시오.

아래 그림과 같이 대상 이미지에서 전 리된 이미지를 볼 수 있습니다.



소스 이미지와 대상 이미지 (EasyImage와 임계값)

8. Open eVision Studio와의 다른 곳에서 대상 이미지를 사용하고 싶으면, 아래 설명한 대로 이미지를 저장 하십시오.

이미지 저장하기

1. 저장하려는 이미지를 클릭하면 해당 이 활성화됩니다.
2. 저장 메뉴를 열려면 다음 중 하나를 수행하십시오.
 - 이미지에서 마우스 오른쪽 버튼을 클릭하십시오.
 - 혹은 기본 메뉴> 이미지를 여십시오.
3. 다른 이름으로 저장...을 클릭하십시오.
4. 파일 형식을 선택 하십시오.(JPEG, JPEG2000, PNG, TIFF 또는 비트맵)
5. 파일 이름을 적고 경로를 선택 하십시오.
6. 저장 버튼을 클릭하십시오.

2. Tutorials

2.1. Preliminaries

- About the application basic interface, operations and workflow, please refer to "[Getting Started](#)" on page 1, available through the **Help** menu. About the **Open eVision** functional concepts, and about the API classes, methods, properties, refer to the main section "[Using Open eVision](#)" on page 1.
- There are often many different ways to get the same results through the graphical interface. The tutorials show only one way for each case.
- All the images used in the tutorials are located by default in the following subdirectory. As this root path is constant, it is not repeated in each tutorial.
C:\Users\Public\Documents\Euresys\Open eVision xx.xx\Sample Images
- By default, in the main toolbar, the **Activate/Deactivate script code generation** toggle button is on. This will insert automatically into the script window the code corresponding to the actions that are taken through the graphical interface. But note that the automatic operations are not inserted into the script until they have been explicitly performed by clicking a dialog box button (**Execute**, **Find**, **Learn...**).

2.2. EasyImage

Converting a Gray-Level Image into a Binary Image

Thresholding

[Single Thresholding](#) - [Double Thresholding](#) - [Histogram-Based Single Thresholding](#) - [Histogram-Based Double Thresholding](#)

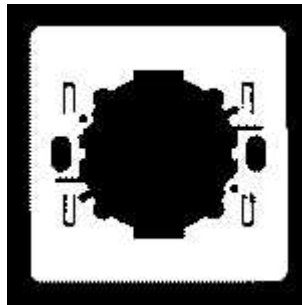
오브젝티브

Following this tutorial, you will learn how to use EasyImage to convert a gray-level source image into a binary destination image. Thresholding an image transforms all the gray pixels into black or white pixels, depending on whether they are below or above a specified threshold. Thresholding an image makes further analysis easier.

You'll need first to load an image (step 1). Then you'll set the thresholding parameters (step 2), and perform the conversion (step 3).



Gray-level source image



Black and white destination image, after thresholding

Step 1: Load the source image

1. From the main menu, click **EasyImage**, then **Threshold**.
2. In the **Simple Threshold** tab, click the **Open** icon of the Source Image area, and load the image file EasyMatch\Switch1.tif.
3. Keep the default variable name for the new Image object, and click **OK**.

Step 2: Set the thresholding parameters

1. In the right area of the **Threshold** dialog box, move the slider to change the threshold, and see directly in the source image a preview of the result.
2. Select the **Minimum residue** option to set a pre-defined algorithm that finds automatically the right threshold.

Step 3: Perform the conversion

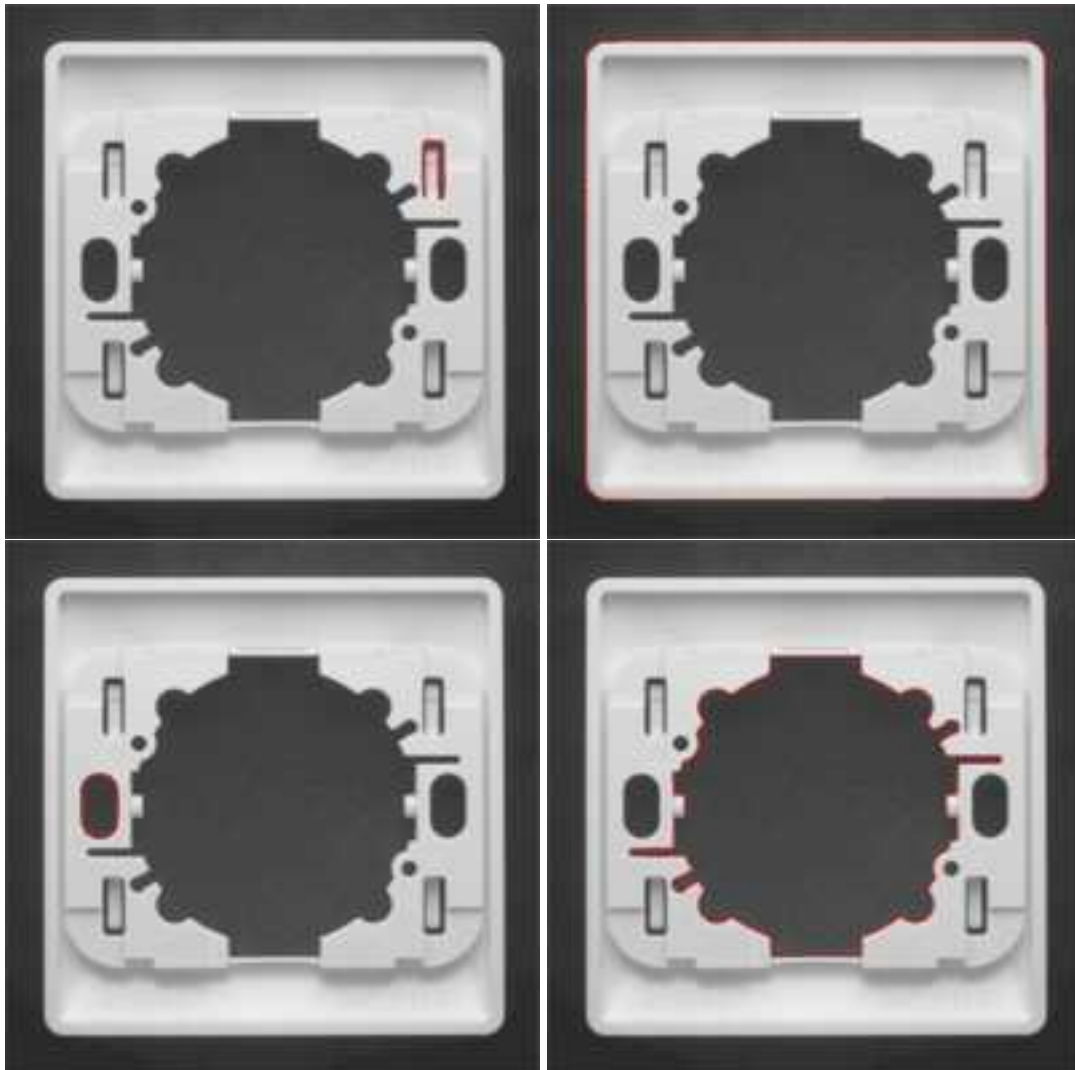
1. Click the **New** icon in the **Destination Image** area to create a new destination image.
2. Keep the default settings for the new Image object, and click **OK**.
3. In the **Threshold** dialog box, click **Execute** to perform the thresholding in the destination image.

Extracting an Object Contour

오브젝티브

Following this tutorial, you will learn how to use EasyImage to trace an object outline in a gray-level image. The contour extraction allows you to get in a path vector all the points that constitute an object contour, just by clicking an edge of this object.

You'll need first to load an image (step 1) and set a vector that will contain all the contour points (step 2). Then you'll click an object edge, and the contour will be extracted automatically (step 3).



Contours are extracted from object edges

Step 1: Load the source image

1. From the main menu, click **EasyImage**, then **Contour**.
2. Click the **Open** icon of the Source Image area, and load the image file EasyMatch\Switch1.tif.
3. Keep the default variable name for the new image object, and click **OK**.

Step 2: Set the destination vector

1. Click the **New** icon in the Destination Vector area.
2. Keep the default settings and variable name for the new vector object.

3. **Ok**를 클릭하십시오.

Step 3: Extract the contour

1. When moving the cursor above the image, an arrow appears.
2. Move the arrow above an object edge, and click.

From this object edge, a contour is traced, and a red line appears around the object.

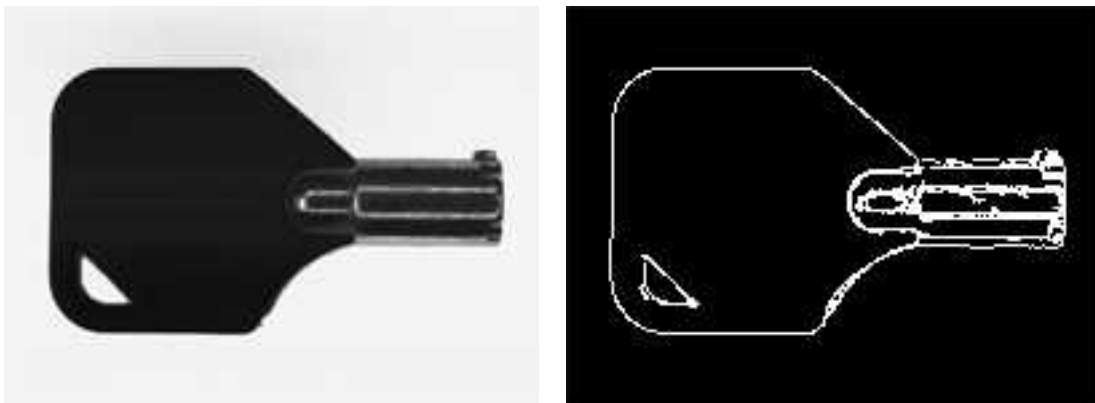
The destination vector is filled with the points constituting the contour.

Transforming a Gray-Level image into its Black and White Edges

오브젝티브

Following this tutorial, you will learn how to use EasyImage to transform a gray-level image to a binary image, keeping only the edges detected in the image. The conversion uses the Canny edge detector algorithm.

You'll need to load a source image (step 1), and simply apply the Canny edge transformation (step 2).



Source image (left) and destination image after Canny edge transformation (right)

Step 1: Load the source image

1. From the main menu, click **EasyImage**, then **Canny Edge Detector**.
2. Keep the default variable name, and click **OK**.
3. Click the **Open** icon of the Source Image area, and load the image file EasyImage\Key1.tif.
4. Keep the default variable name, and click **OK**.

Step 2: Apply the Canny edge transformation

1. Click the **New** icon in the Destination Image area to create a new destination image.
2. Keep the default settings, and click **OK**.

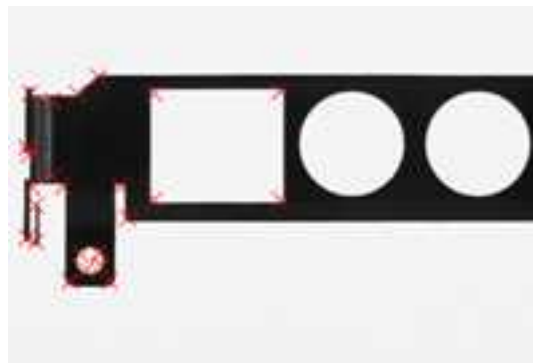
3. In the canny edge detector dialog box, click **Apply** to perform the operation in the destination image.

Detecting the Corners of an Object Using Harris Corner Detector

오브젝티브

Following this tutorial, you will learn how to use EasyImage to detect the corners of an object. The detection uses the Harris corner detector algorithm.

You'll need to load a source image (step 1), and simply apply the Harris corner detection (step 2).



Corners are detected in the source image

Step 1: Load the source image

1. From the main menu, click **EasyImage**, then **Harris Corner Detector**.
2. Keep the default variable name, and click **OK**.
3. Click the **Open** icon of the Source Image area, and load the image file EasyGauge\Bracket1.tif.
4. Keep the default variable name, and click **OK**.

Step 2: Apply the Harris corner detection

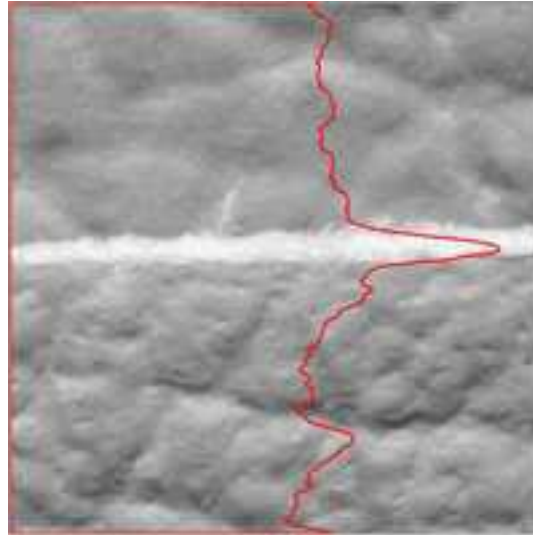
1. In the Harris corner detector dialog box, enter 2.3 for the Scale property.
2. Click **Apply** to perform corners detection.
3. Click **Results** to display the coordinates of all detected corners.
4. The **Columns** button allows you to display additional properties in the results list.

Detecting a Horizontal or Vertical Line Using Projection

오브젝티브

Following this tutorial, you will learn how to use EasyImage to detect defects (horizontal/vertical line) in a gray-level image.

You'll need first to load a source image (step 1), set a vector (step 2), and then detect the defect (horizontal line) (step 3).



Defects can be detected using the image projection

Step 1: Load the source image

1. From the main menu, click **EasyImage**, then **Projection**.
2. Click the **Open** icon of the Source Image area, and load the image file EasyImage\Leather.bmp.
3. Keep the default variable name for the new image object, and click **OK**.

Step 2: Set the destination vector

1. Click the **New** icon in the Destination Vector area.
2. Select the BW32 option for the vector type, and click **OK**.

Step 3: Detect the defects

1. In the Image Projection dialog box, select the **column** button, and click **Execute** to perform the operation.
2. The resulting vector and the corresponding plot are displayed in the destination vector window. The graphical result also appears on the image. Each vector value is the sum of all pixels values across the corresponding horizontal row (or vertical column). By this mean, horizontal (or vertical) defects are easily detected.

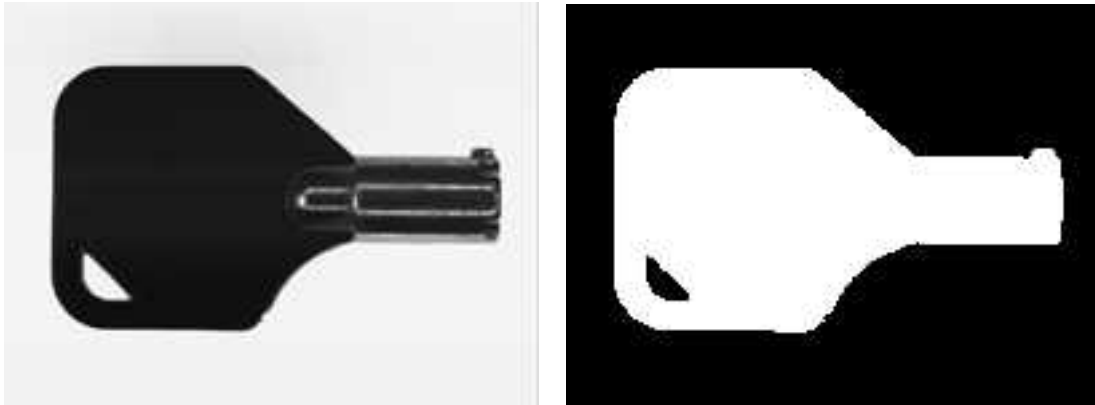
Creating a Flexible Mask

Using Flexible Masks

오브젝티브

Following this tutorial, you will learn how to create a flexible mask from a source image, to restrict a future processing to an arbitrary-shaped do-care area.

Flexible masks can be created in any ways to build a bi-level image. Here, we will first load the source image (step 1), and then successively invert it, and threshold it (steps 2-3). The resulting image —the flexible mask— will be saved as a new image file (step 4). This new image file is a bi-level image. However, there are still black areas that need to be erased, before using the image as a flexible mask. You can use a third-party software, such as Paint, to clear the unwanted areas.



Source image (left) and flexible mask image (right)

Step 1: Load the source image

1. From the main menu, click **EasyImage**, then **Arithmetic & logic**.
2. Click the **Open** icon of the Source Image 0 area, and load the image file EasyImage\Key1.tif.
3. Keep the default variable name for the new image object, and click **OK**.

Step 2: Invert the image

1. Click the **New** icon of the **Destination** area.
2. Keep the default settings for the new image object, and click **OK**.
3. In the **Operation** drop-down list, select **Invert**, and click **Execute**.

Step 3: Threshold the image

1. From the main menu, click **EasyImage**, then **Threshold**.
2. In the Source Image area, select the inverted image from the drop-down list.
3. Click the **New** icon of the Destination area.
4. Keep the default settings for the new image object, and click **OK**.
5. Select the Absolute option, enter '46' as the threshold value, and click **Execute**.

Step 4: Save the flexible mask

1. Right-click in the destination image, and select **Save As...**.
2. Type a file name for the new flexible mask file. Finally, click **Save**.

- ✓ The new image now is a bi-level image. However, there are still black areas that need to be erased, before using the image as a flexible mask. You can use a third-party software, such as Paint, to clear the unwanted areas.

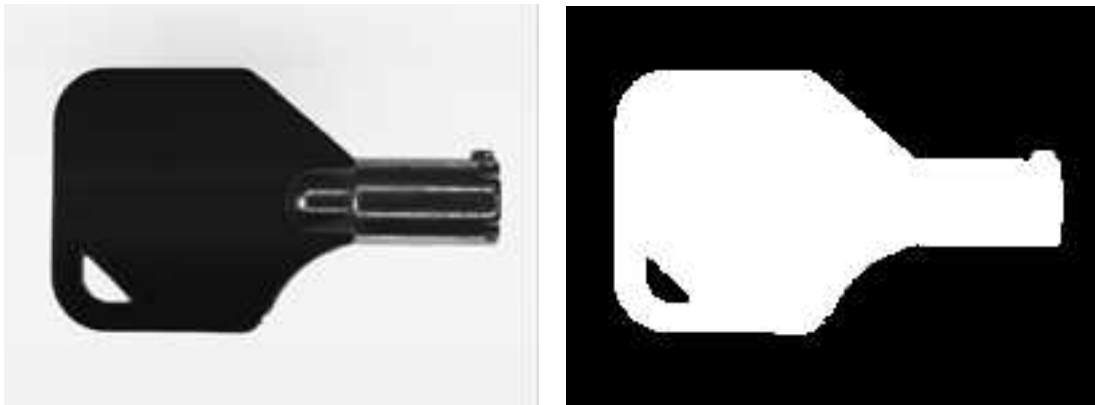
Computing Gray-Level Statistics Using a Flexible Mask

Using Flexible Masks

오브젝티브

Following this tutorial, you will learn how to compute gray-level statistics on an arbitrary-shaped area only.

You'll need first to load a source image (step 1), and a flexible mask image (step 2). The mask image must be applied on the source image (step 3), to separate do-care areas (that must be considered) and don't-care areas (that should not be considered). Finally, the gray-level statistics are computed on the do-care area only (step 4).



Source image (left) and flexible mask image (right)

Step 1: Load the source image

1. From the main menu, click **EasyImage**, then **Image Statistics**, **Gray Scale**.
2. Click the **Open** icon of the Source Image area, and load the image file EasyImage\Key1.tif.
3. Keep the default variable name for the new image object, and click **OK**.

Step 2: Load the flexible mask image

1. From the main menu, click **Image**, then **Open...**
2. Load the image file EasyImage\Mask2.bmp.
3. Keep the default variable name for the new image, and click **OK**.

Step 3: Apply the flexible mask on the source image

1. In the Mask area of the **Gray Scale Image Pixels Statistics** dialog box, select the mask image from the drop-down list.
2. The source image preview in the dialog box shows (in red diagonal lines) the don't-care area, that is the area that will be not be considered when computing the gray-level statistics.

Step 4: Compute the gray-level statistics

1. Select the **Pixel Count** check-box.
2. Click **Execute**.

The results are displayed in the read-only fields below.

Detecting the Corners of an Object Using Hit-and-Miss Transform

Hit-and-Miss Transform

오브젝티브

Following this tutorial, you will learn how to use EasyImage to detect top corners in an image, using the hit-and-miss transform.

You'll need to load a source image (step 1), set the kernel that represents a top corner (step 2), and then set a destination image and simply execute the hit-and-miss transform (step 3).



Source image (left) and top corner detected in the source image (white dot)

Step 1: Load the source image

1. From the main menu, click **EasyImage**, then **Hit And Miss**.
2. Click the **Open** icon of the Source Image area, and load the image file EasyImage\Diamond.bmp.
3. Keep the default variable name, and click **OK**.

Step 2: Set the hit-and-miss kernel

- In the **Hit And Miss** dialog box, set the kernel according to the following values:

	-1	0	1
-1	Don'tCare	Background	Don'tCare
0	Background	Foreground	Background
1	Foreground	Foreground	Foreground

Kernel that detects top corners

Step 3: Apply the hit-and-miss transform

1. Click the **New** icon of the **Destination Image** area.
2. Keep the default parameters and variable name, and click **OK**.
3. Click **Execute** to perform the operation.
The top corner (white dot) is detected.
4. Try with other kernel configurations to detect the other corners.

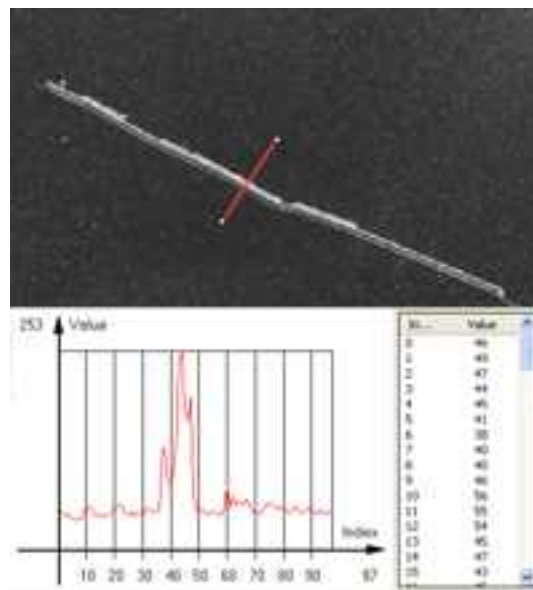
Extracting a Vector Using Profile Function

Profile Sampling

오브젝티브

Following this tutorial, you will learn how to use EasyImage to detect scratches.

You'll need first to load an image (step 1), set a destination vector, and detect the scratches (step 2).



Scratches can be detected using a profile

Step 1: Load the source image

From the main menu, click **EasyImage**, then **Profile**.

1. Click the **Open** icon of the **Source Image** area, and load the image file EasyImage\Plastic.tif.
2. Keep the default variable name for the new image object, and click **OK**.

Step 2: Set the destination vector and detecting the scratches

1. Click the **New** icon in the Destination Vector area.
2. Select the BW8 option for the vector type, and click **OK**.

3. A profile appears on the image (red line segment). In the destination vector window, vector values correspond to pixels along the line segment.

The scratch is detected as a sharp deviation in the vector graph.

4. Using the mouse, drag the handles to move or resize the red line segment, and observe the plot evolution.

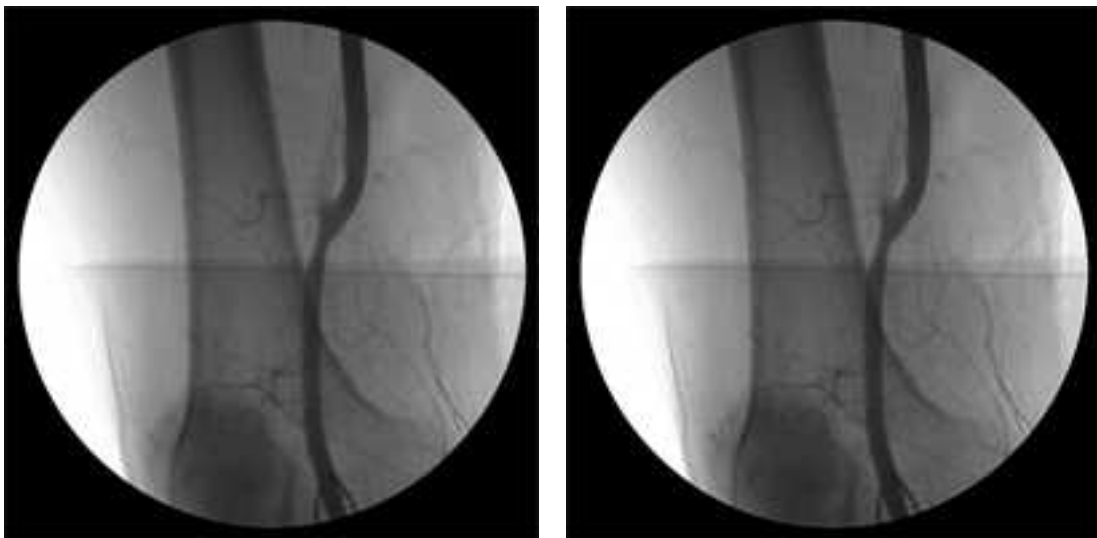
The sharp deviation appears whenever the line segment is placed across the scratch.

Enhancing an X-ray image

오브젝티브

Following this tutorial, you will learn how to use EasyImage to enhance an X-ray image.

You'll need first to load an image (step 1), then define convolution parameters to enhance the image (step 2).



Source image (left) and enhanced image, after predefined and user-defined convolutions (right)

Step 1: Load the source image

1. From the main menu, click **EasyImage**, then **Convolution**.
2. Click the **Open** icon of the **Source Image** area, and load the image file EasyImage\XRay.bmp.
3. Keep the default variable name for the new image object, and click **OK**.
4. Click the **New** icon of the **Destination Image** area to create a new destination image.
5. Keep the default variable name and click **OK**.

Step 2: Set the convolution parameters

1. From the **Predefined kernels** drop-down list, select **Highpass2**, and click **Execute** to perform the operation.

The image is no longer blurred but the result is bad because the filter has revealed the noise of the source image. We need to create a new convolution kernel that will apply a softer high-pass filtering.

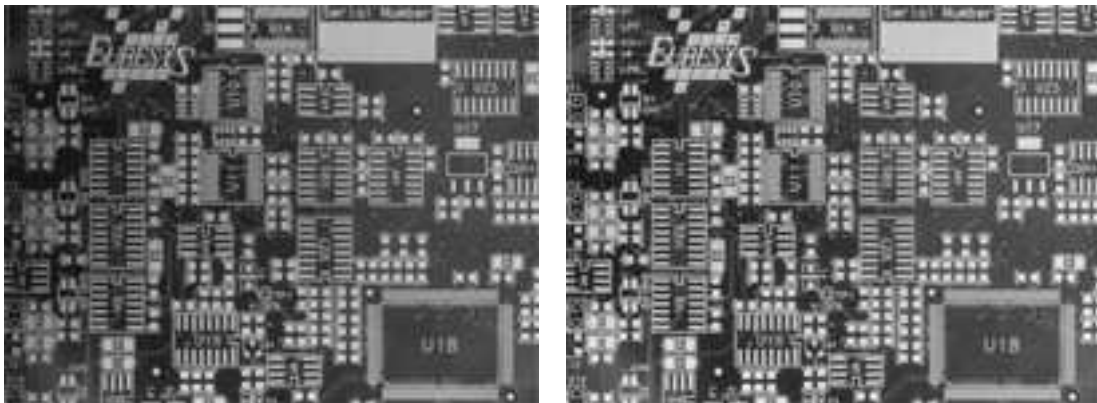
2. Click the **New** icon next to the **User defined kernels** drop-down list.
3. Keep the default dimension (3x3) and variable name, and click **OK**.
4. Enter the following kernel data from left to right and top to bottom: -1, -1, -1; -1, 15, -1; -1, -1, -1, and click **Apply**.
5. Click **Execute** in the Convolution dialog box to perform the operation. The image is much clearer now.

Correcting Non-Uniform Illumination

오브젝티브

Following this tutorial, you will learn how to use EasyImage to correct non-uniform illumination in an image.

You'll need first to load an image (step 1), load a light reference image (step 2), and perform the correction (step 3).



Source image, with non-uniform illumination (left) and corrected image (right)

Step 1: Load the source image

1. From the main menu, click **EasyImage**, then **Uniformize**.
2. Click the **Open** icon of the Source Image area, and load the image file EasyImage\Board (original).tif.
3. Keep the default variable name for the new image object, and click **OK**.

Step 2: Load the reference image

1. Click the **Open** icon of the **Light Reference** area, and load the image file EasyImage\Board (light reference).tif.

To obtain the light reference image, we used a white screen illuminated in the same condition as the board (original image).

2. Keep the default variable name for the new image object, and click **OK**.

Step 3: Perform the correction

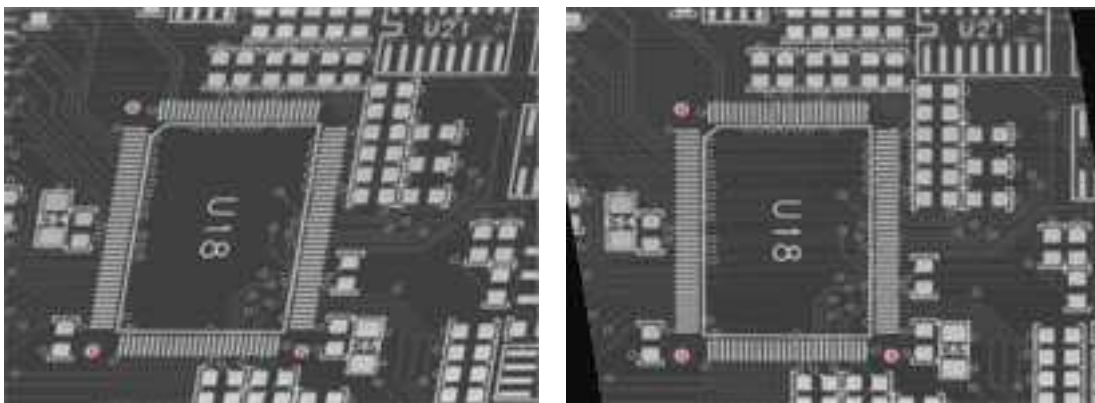
1. Click the **New** icon in the **Destination Image** area to create a new destination image.
2. Keep the default values and click **OK**.
3. Click **Execute** to perform the operation.
4. In both source and destination images, right-click and select **3D Rendering**.
5. In the new 3D windows, click and drag the mouse to rotate the view. Compare the profiles.

Correcting Shear Effect

오브젝티브

Following this tutorial, you will learn how to use EasyImage to correct a shear effect in an image. The following image is taken by a line-scan camera. The camera sensor was misaligned, resulting in a so-called shear effect.

You'll need first to load an image (step 1), create a destination image (step 2), and then set pivots parameters to perform the correction (step 3).



Source image, with a shear effect (left) and corrected image (right)

Step 1: Load the source image

1. From the main menu, click **EasyImage**, then **Register**.
2. Click the **Open** icon of the Source Image area, and load the image file EasyImage\Shear.tif.
3. Keep the default variable name for the new image object, and click **OK**. Three pivots points appear in the image.

Step 2: Create a destination image

1. Click the **New** icon of the **Destination Image** area.
2. Enter '768' and '576' as image width and height, and click **OK** to accept the default name. Three pivots points appear in the image.

Step 3: Set the pivots parameters

1. In the source image, using the mouse, drag each pivot to the center of the fiducial marks (the dots around the U18 area).

Notice that the source pivots coordinates, in the Register dialog box, have changed accordingly.

2. To correct the image, enter the following destination pivots coordinates:

- ☐ X0: 170
- ☐ Y0: 495
- ☐ X1: 470
- ☐ Y1: 495
- ☐ X2: 170
- ☐ Y2: 144

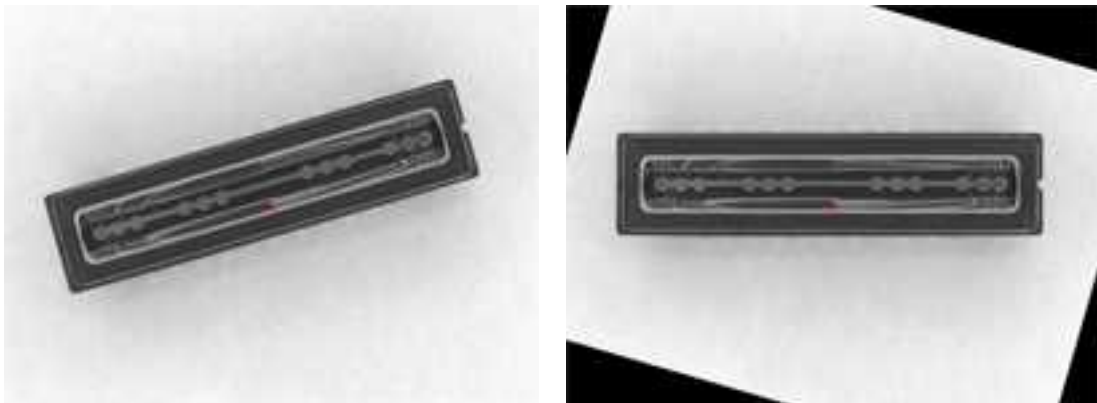
3. Click **Execute** to perform the operation.

Correcting Skew Effect

오브젝티브

Following this tutorial, you will learn how to use EasyImage to correct skew effect in an image.

You'll need first to load an image (step 1), create a destination image (step 2), and then set a correction angle to perform the correction (step 3).



Source image, with a skew effect (left) and corrected image (right)

Step 1: Loading the source image

1. From the main menu, click **EasyImage**, then **Scale and Rotate**.
2. Click the **Open** icon of the **Source Image** area, and load the image file EasyImage\CCD.tif.
3. Keep the default variable name for the new image object, and click **OK**.

Step 2: Creating a destination image

1. Click the **New** icon of the **Destination Image** area.
2. Enter '768' and '576' as image width and height, and click **OK** to accept the default name.

Step 3: Setting the correction angle

1. Select the **Rotate** option, and enter -16.17 in the Angle (Deg) field. (To measure this rotation angle, refer to Measuring the rotation angle of an object.)
2. From the Interpolation bits drop-down list, select 8 bits to get a better result.
3. Click **Execute** to perform the operation.

2.3. EasyColor

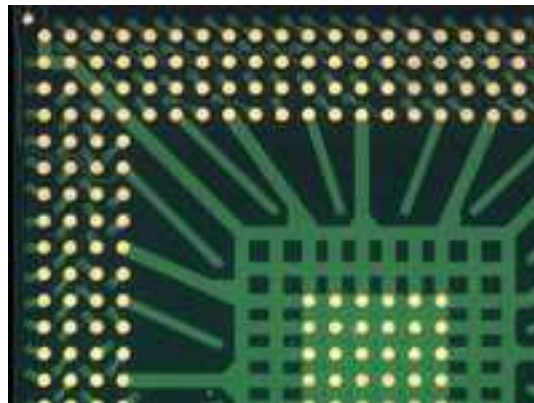
Performing Thresholding on Color Images

Color Components

오브젝티브

Following this tutorial, you will learn how to use EasyColor to segment a color source image, by setting a threshold value for each color component of the current color system. For example, to retrieve the solder pads on a PCB, you'll perform a color segmentation based on the golden pixels (H), with a loose discrimination on the brightness (L) and saturation (S), to eliminate surface and lighting effects.

You'll need first to load a color source image, create a destination image, and a color lookup table (steps 1-3). Then, you'll set the color system and tune each component tolerance to get a satisfying segmentation of the solder pads (step 4).



소스 이미지



임계값 적용 이미지

Step 1: Load the source image

1. From the main menu, click **EasyColor**, then **Threshold**.
2. Click the **Open** icon of the **Source Image** area, and load the image file EasyColor\BGA Substrate Color.jpg.
3. Keep the default variable name for the new Image object, and click **OK**.
4. Disable the **Preview Mode** check-box to see the raw source image.

Step 2: Create a destination image

1. Click the **New** icon of the **Destination Image** area.
2. Keep the default variable name for the new Image object, and click **OK**.

Step 3: Create a color lookup table

1. Click the **New** icon of the **Color Lookup** area.
2. Keep the default variable name for the new color lookup object, and click **OK**.

Step 4: Perform the color segmentation

1. Select **LSH** from the **Color System** drop-down list.
2. In the source image, click in a golden pad. The pixel lightness, saturation and hue values are updated in the **Color Threshold** dialog box.
3. Adjust the tolerance of lightness and saturation to enlarge the range of thresholded pixels, until you get a satisfying segmentation in the destination image.

Performing Color Segmentation

Color Components

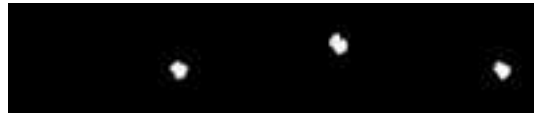
오브젝티브

Following this tutorial, you will learn how to use EasyImage to perform color segmentation.

You'll need first to load an image (step 1), create a color lookup table (step 2), and perform the segmentation (step 3).



소스 이미지



Segmented image

Step 1: Load the source image

1. From the main menu, click **EasyColor**, then **Threshold**.
2. Click the **Open** icon of the Source Image area, and load the image file EasyColor\Pills.tif.
3. Keep the default variable name for the new image object, and click **OK**.
4. Disable the **Preview Mode** check-box to see the raw source image.

Step 2: Create a color lookup table

1. Click the **New** icon of the Color Lookup area.
2. Keep the default variable name for the new color lookup object, and click **OK**.

Step 3: Perform the color segmentation

1. Select **LSH** from the **Color System** drop-down list.
2. In the source image, click in the center of a green pill. The pixel lightness, saturation and hue values are updated in the **Color Threshold** dialog box.
3. Increase the lightness tolerance up to 120. Increase the saturation tolerance up to 50.
4. Enable the **Preview Mode** check-box to see the result of the segmentation. If needed, click in the green pills to improve the result.
5. Click the **New** icon of the Destination Image area.
6. Keep the default settings for the new Image object, and click **OK**.
7. The new image is automatically thresholded. Clicking **Execute** will insert the corresponding code into the script windows.

2.4. EasyObject

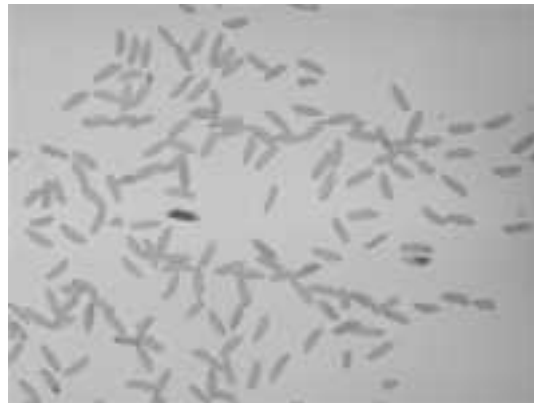
Removing Non-Significant Objects After Image Segmentation

Image Segmenter

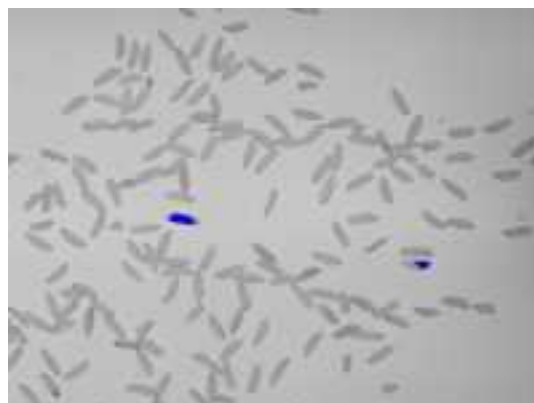
오브젝티브

Following this tutorial, you will learn how to use EasyObject to detect bad rice grains (largely dark) among many normal rice grains (largely light).

You'll need first to load the source image (step 1). Then you'll perform the image segmentation, based on a threshold value (step 2). All the detected objects are dark, but some are too small to be significant. So, you'll set a minimum object area (number of pixels), and remove the smallest objects (step 3). Finally, you'll get only the objects that really represent bad rice grains.



소스 이미지



Bad rice grains are detected

Step 1: Load the source image

1. From the main menu, click **EasyObject**, then **New EasyObject Tool**.
2. Keep the default variable name for the new object, and click **OK**.

3. In the **Encoder** tab, click the **Open** icon of the Source Image area, and load the image file EasyObject\Rice.jpg.
4. Keep the default variable name for the new image, and click **OK**.

Step 2: Perform image segmentation

1. In the **Encoder** tab, select the **Black Layer** check-box, and unselect the **White Layer** check-box.
2. Click the **...** button around the **Threshold** field. In the Threshold dialog box, select **Absolute**, enter '115', and click **OK**.
3. Click **Encode** to detect the dark objects. In the image, each object is drawn using a different color.
4. Click **Results** to display the list of all the detected objects. Clicking an object in the image highlights it in the list, and vice versa.

Step 3: Remove the smallest objects

1. In the objects list, click **Columns**.
2. Tick the **Area** check-box, and click **OK**. In the list view, a new Area column appears, displaying each object area.
3. Click the Area column header to sort the objects. There are many small objects (area < 100) that may be considered as noise.
4. In the **Selection** tab, select **Area** from the Feature drop-down list. Select **'Less'** from the Mode drop-down list. In the Threshold field, enter '100'.
5. Click **Remove**. All the objects with an area smaller than 100 pixels have been removed from the list. The two remaining objects are the bad rice grains.

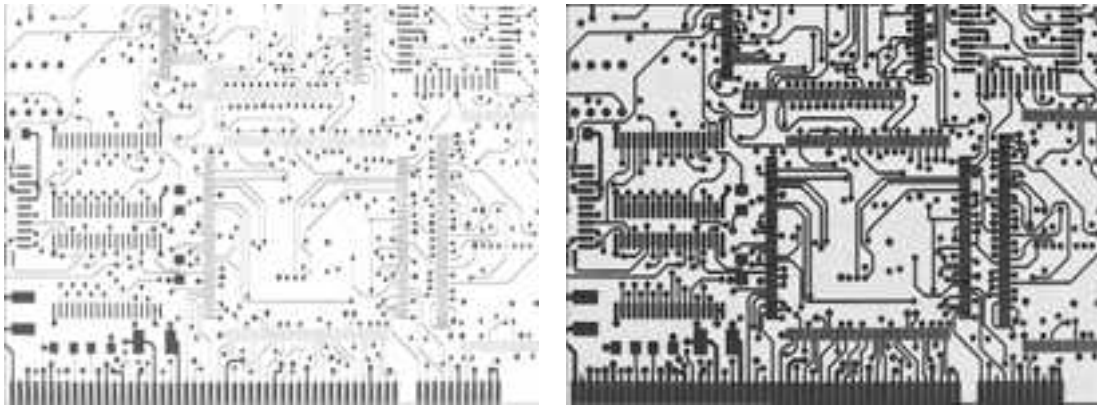
Detecting Differences Between Images Using Min-Max References

Selecting and Sorting Blobs

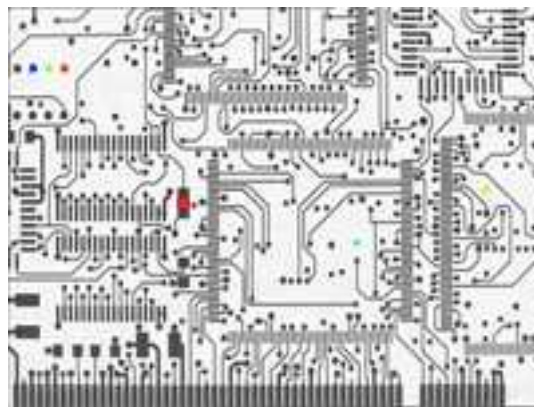
오브젝티브

Following this tutorial, you will learn how to use EasyObject to compare images. In this example, we will check the quality of a PCB film.

You'll need first to load a reliable source image (step 1), from which two reference images (min and max) will be built (step 2). Then you'll load another image to be inspected (step 3), and perform the comparison with the min and max reference images (step 4). The differences will be detected.



High (left) and low (right) threshold reference images



In another image, differences are detected

Step 1: Load the source image

1. From the main menu, click **EasyObject**, then **Make Min Max**.
2. Click the **Open** icon of the Source Image area, and load the image file EasyObject\FilmOk.png.
3. Enter 'filmOk' for the name of the new image, and click **OK**.

Step 2: Build min and max reference images

1. Click **Execute**. Min and Max reference images are created, based on the source image.
 - filmOkMax is computed by dilating filmOk a given number of times ('geometric margin') and adding a constant ('gray level margin') to every pixel. filmOkMin is computed by eroding filmOk the same number of times and subtracting the same constant to every pixel.
 - The geometric margin can be seen as a position tolerance between the image to be inspected and the reference.
 - The gray level margin introduces a tolerance to lighting variations.

Step 3: Load an image to be inspected

1. From the main menu, click **EasyObject**, then **New EasyObject Tool**.
2. Keep the default variable name for the new object, and click **OK**.

3. In the **Encoder** tab, click the **Open** icon of the Source Image area, and load the image file EasyObject\FilmBad.png.
4. Enter 'filmBad' for the name of the new image, and click **OK**.

Step 4: Compare the image with the reference images

1. In the **Encoder** tab, select **ImageRange** in the segmentation method drop-down list.
2. Disable the **White Layer** check-box, and enable the **Black Layer** check-box.
3. Click the **...** button around the High Image field. Select **filmOkMax** in the drop-down list, and click **OK**.
4. Click the **...** button around the Low Image field. Select **filmOkMin** in the drop-down list, and click **OK**.
5. Click **Encode**. Eight differences are highlighted in the image.
6. Click **Results** to get further information about the detected objects. They may be further filtered and analyzed using object features selection and sorting capabilities of EasyObject.

Detecting Printing Errors Using a Flexible Mask

Generating a Flexible Mask from an Encoded Image

오브젝티브

Following this tutorial, you will learn how to use a flexible mask to target and search specific areas in the image.

You'll need first to load a source image (step 1), and a flexible mask image (step 2), that can be automatically applied on the source image to separate do-care areas (that must be considered) and don't-care areas (that should not be considered). Then, you'll perform the inspection only on do-care areas (step 3).



소스 이미지



Mask image



결과

Step 1: Load the source image

1. From the main menu, click **EasyObject**, then **New EasyObject Tool**.
2. Keep the default variable name for the new object, and click **OK**.
3. In the **Encoder** tab, click the **Open** icon, and load the image file EasyObject\Mobile3.jpg.
4. Keep the default variable name for the new image, and click **OK**.

Step 2: Load the flexible mask image

1. In the **Encoder** tab, click the **Open** icon, and load the flexible mask image file EasyObject\MobileMask.bmp.
2. Enter 'mask' for the name of the new image, and click **OK**. In the **Mask** area of the **Encoder** tab, notice that the mask image is selected from the drop-down list: the mask is automatically applied on the source image, because its name contains 'mask', and because it has been loaded from the coded image dialog box. The source image preview in the dialog box shows (in red diagonal lines) the don't-care area, that is the area that will be not be considered when encoding the source image.

Step 3: Inspect the image

1. In the **Segmentation** area of the **Encoder** tab, click the **...** button to display the threshold dialog box.

2. Select Absolute and enter 202 for the threshold. Click **OK** to close the dialog box.
3. Click **Encode** to locate the objects (in the do-care areas only). In the source image, each object is drawn using a different color. Three printing errors can be observed:
 - ☐ The digit '7' is partially printed.
 - ☐ The '+' sign is missing.
 - ☐ The handset is printed on a lighter tone.
4. Click Results to display the statistics on each object.
 - ☐ Selecting an object in the list highlights it in the image.
 - ☐ Selecting **Columns** and **Drawing** will display more options.

2.5. EasyMatch

Learning a Pattern and Creating an EasyMatch Model File

Pattern Learning

오브젝티브

Following this tutorial, you will learn how to use EasyMatch to learn a model from a reference image, and save it as an EasyMatch model file.

You'll need first to load the reference image (step 1). Then, you'll learn it as the reference model (step 2). And you'll save the model as an EasyMatch model file (step 3).



Reference image

Step 1: Load the reference image

1. From the main menu, click **EasyMatch**, then **New Match Tool**.
2. Keep the default variable name for the new matcher object, and click **OK**.
3. In the **Learning** tab, click the **Open** icon, and load the image file EasyMatch\FramModel.tif.
4. Keep the default variable name for the new image object, and click **OK**.

Step 2: Learn the reference image

- In the **Learning** tab, click **Learn** to acquire the model pattern.

Step 3: Save the model file

1. In the **Learning** tab, click the **Save As...** button.
2. Type a file name for the new EasyMatch model file. Its extension will be .mch.
3. **저장을** 클릭합니다.

Matching a Pattern According to a Model File

Pattern Matching and Retrieving Results

오브젝티브

Following this tutorial, you will learn how to use EasyMatch to load an EasyMatch model file, and search for occurrences of the pattern in an image.

You'll need first to load the model file, and a source image where the model will be searched for (steps 1-2). Then the pattern matching is fully automatic (step 3).



Occurrences of the model are automatically highlighted

Step 1: Load the model file

1. From the main menu, click **EasyMatch**, then **New Match Tool**.
2. Keep the default variable name for the new matcher object, and click **OK**.
3. In the **Learning** tab, click **Load...** to open the model file EasyMatch\Switch.MCH. The model contains all necessary information about the pattern we are searching for.

Step 2: Load a source image

1. In the **Matching** tab, click the **Open** icon, and load the image file EasyMatch\Switch1.tif.
2. Keep the default variable name for the new image object, and click **OK**.

Step 3: Perform the pattern matching

1. The pattern matching is automatically performed on the source image, and the matching occurrences are highlighted. Clicking **Execute** will insert the corresponding code into the script windows.
2. Further information about each occurrence can be found by clicking **Results**.
3. Click in a row to see the corresponding occurrence highlighted in the image.

Learning a Pattern According to an ROI

Pattern Learning

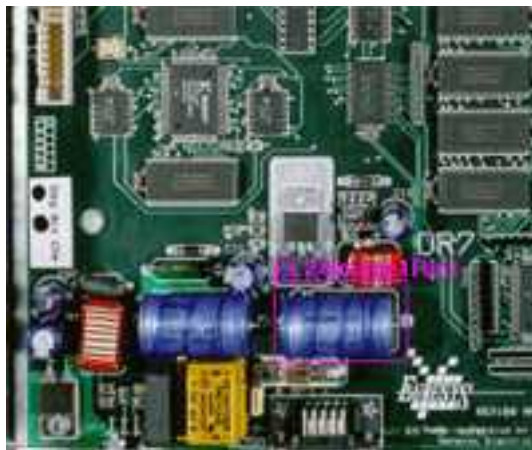
오브젝티브

Following this tutorial, you will learn how to use EasyMatch to learn a model from an ROI in a source image, and to perform pattern matching on the same image.

You'll need first to load the source image, and define an ROI inside (steps 1-2). Then, you'll have to learn the model, using this ROI (step 3). Finally, you'll perform pattern matching in the source image (step 4), and will find additional occurrences of the model.



ROI that will be learned



Occurrences matching the model ROI

Step 1: Load the source image

1. From the main menu, click **Image**, then **Open**.
2. Load the image file EasyMatch\BOARD.JPG.
3. Keep the default variable name for the new image object, and click **OK**.

Step 2: Define an ROI

1. Right-click in the image, and select **New ROI...** from the contextual menu.
2. Keep the default variable name for the new ROI object, and click **OK**. A default ROI is placed over the image (blue rectangle with handles).

The ROI management dialog box is opened.

3. Resize the ROI and move it around one of the blue capacitors at the lower left part of the image.

Step 3: Learn a model from the ROI

1. From the main menu, click **EasyMatch**, then **New Match Tool**.
2. Keep the default variable name for the new matcher object, and click **OK**.
3. In the **Learning** tab of the matcher dialog box, select the ROI object from the Source Image drop-down list, and click **Learn** to acquire the model pattern.

Step 4: Match the pattern

1. In the **Matching** tab, increase the **Max Occurrences** field to 2.
2. Select the image object from the Source Image drop-down list.
3. Click **Execute**. The occurrences of the learned model are highlighted in the source image.
4. Further information about each occurrence can be found by clicking **Results**.
5. Click in a row to see the corresponding occurrence highlighted in the image.

Improving the Score of Matching Instances by Using "Don't Care Areas"

Setting Search Parameters

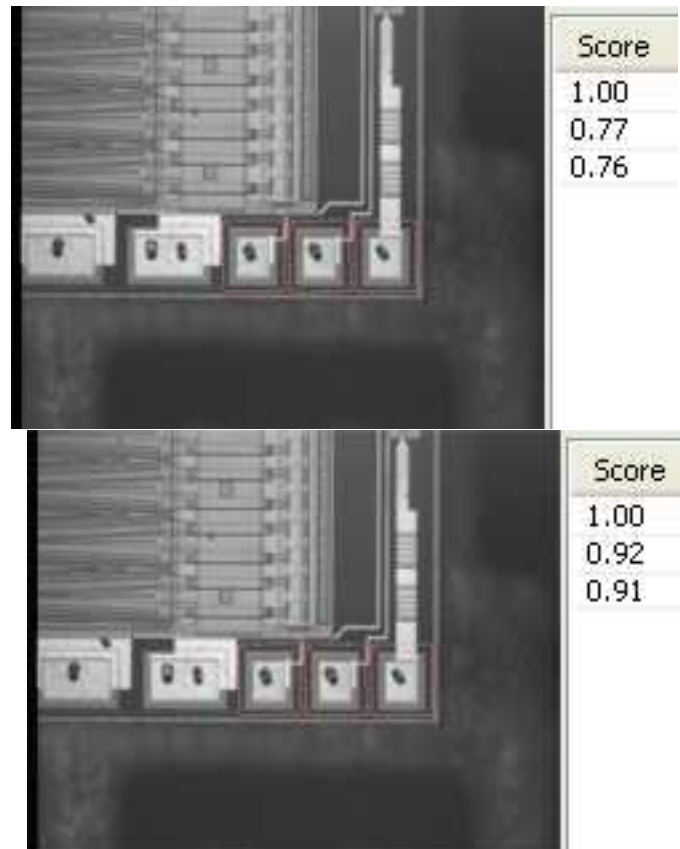
오브젝티브

Following this tutorial, you will learn how to use EasyMatch to handle "don't care areas". "Don't care areas" help to define in the image the meaningful features only, by masking the areas that might change from image to image.

You'll need first to load a reference image and learn the reference model (steps 1-2). Then you'll perform automatic pattern matching of instances of the reference model, without using "don't care areas" (step 3). As the matching scores of the found instances are not high enough, you'll define a "don't care area" on the reference model, and restart the detection. The matching scores are much better (steps 4-5).



Reference model



Found instances and matching scores, without (left) and with (right) using "don't care areas"

Step 1: Load the reference image

1. From the main menu, click **EasyMatch**, then **New Match Tool**.
2. Keep the default variable name for the new Matcher object, and click **OK**. The Matcher management dialog box is opened.
3. In the **Learning** tab, click the **Open** icon, and load the image file EasyMatch\Die Pad Model 1.bmp.
4. Keep the default variable name for the new Image object, and click **OK**.

Step 2: Learn the reference model

- Click **Learn** to acquire the model pattern.

Step 3: Detect instances of the reference model without "don't care areas"

1. In the **Matching** tab, click the **Open** icon, and load the image file EasyMatch\Die Pad 1.bmp.
2. Keep the default variable name for the new Image object, and click **OK**. An instance matching the reference model is highlighted.
3. Increase the **Max Occurrences** field to 3. Enable the **Sub-Pixel Interpolate** check-box to increase the matching precision.
4. Enter '-10.0' as the Minimum Angle (Deg). (Check that the angle is displayed in degrees. If not, select the angles unit from **View > Option** menu.)
5. Enter '10.0' as the Maximum Angle (Deg).

6. Click **Execute**. The pattern locations are highlighted in the source image.
7. Click **Results** to see the matching score of each found instance. The last two scores are rather bad. This is mainly due to the bright rectangle on the upper part of the reference image we have learned. We can improve the scores by using a "don't care area" to mask this bright area.

Step 4: Define the "don't care area"

1. In the **Don't Care Areas** tab, select the **Rectangle** radio button from the **Blacken Inside** group.
2. In the reference image, move your mouse pointer on the lower left corner of the bright rectangle, left-click and drag the don't care area (rectangle with red stripes) to the upper right corner of the bright rectangle to mask out this area.
3. In the **Don't Care Areas** tab, click **Learn**.

Step 5: Detect instances of the reference model with "don't care areas"

1. In the **Matching** tab, click **Execute**. The instances matching the reference model are still highlighted.
2. Click **Results** to compare the new matching scores. They are much better.

2.6. EasyFind

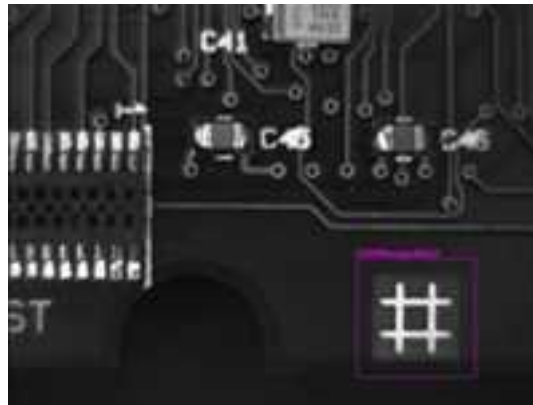
Detecting Highly-Degraded Occurrences of a Reference Model in Multiple Files

Pattern Finding and Retrieving Results

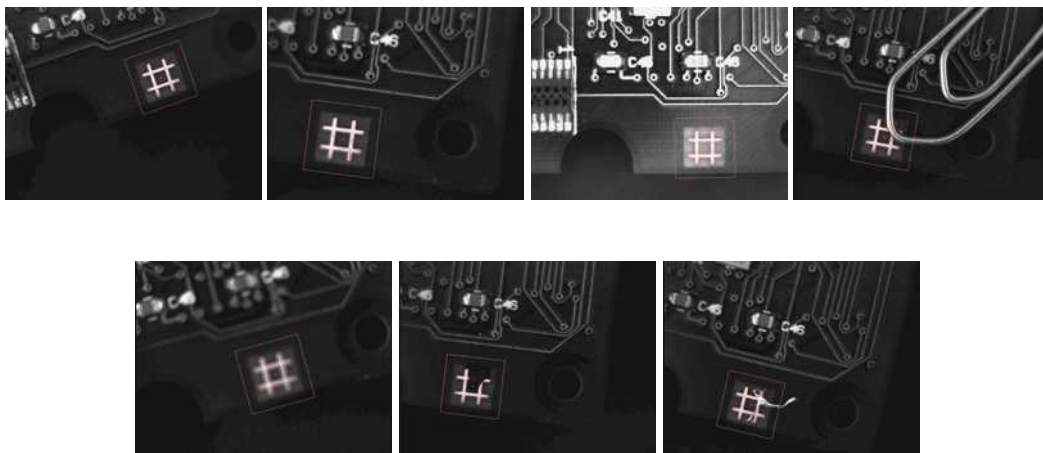
오브젝티브

Following this tutorial, you will learn how to use EasyFind to detect in multiple images highly-degraded occurrences of a reference model. The degradation can be due to noise, blur, occlusion, missing parts or unstable illumination conditions.

You'll need first to load a reference image, define an ROI where EasyFind will learn the reference model, and set rotation and scaling tolerances for the expected occurrences to search for (steps 1-4). Then you're ready to open multiple files, and perform automatic detection of occurrences (even highly-degraded) of the reference model (steps 5-6).



Reference model



Occurrences of the reference model are found, even if highly-degraded

Step 1: Load the reference image

1. From the main menu, click **EasyFind**, then **New EasyFind Tool**.
2. Keep the default variable name for the new PatternFinder object, and click **OK**. The PatternFinder management dialog box is opened.
3. In the **Model** tab, click the **Open** icon, and load the image file EasyFind\Fiducial 1.tif.
4. Keep the default variable name for the new image object, and click **OK**.

Step 2: Create an ROI to define the reference model on the reference image

1. In the image, right-click and select **New ROI...** item from the menu.
2. Keep the default variable name for the new ROI object, and click **OK**. A default ROI is placed over the image (blue rectangle with handles). The ROI management dialog box is opened.
3. Drag the ROI over the reference model and resize it using its handles. Alternatively, enter the following coordinates in the ROI dialog box: 500, 365, 170, 170 for OrgX, OrgY, Width, and Height respectively, and click **Close**.

Step 3: Learn the reference model

1. In the PatternFinder **Model** tab, select the ROI object from the source image drop-down list, and click **Learn**. The reference model is perfectly detected (green edges).
2. In the PatternFinder **Search Field** tab, select the Image object from the source image drop-down list. Tick the Draw Features check-box.

The model location and feature points are highlighted in the source image.

Step 4: Set rotation and scaling tolerances

- In the PatternFinder **Allowances** tab, set both angle tolerance and scale tolerance to 25.0.

Step 5: Select multiple images

1. In the PatternFinder **Search Field** tab, click the **Open** icon. Select the images files EasyFind\Fiducial 2.tif to Fiducial 8.tif (use the shift key to select multiple files), and click **Open**.
2. Keep the default variable name for the new Image object, and click **OK**. The last image appears. The reference model is found, even if highly-degraded.
3. Detection of the reference model is automatically performed. It is not necessary to click Find once a new image appears, as inspection is automatically performed. However, clicking **Find** will insert the corresponding code into the script windows.
4. Click **Results** to find more information about the found instance.

Step 6: Browse multiple images

- In the PatternFinder **Search Field** tab, click the **Load Next** or **Load Previous** icons.

The image files appear, and the reference model is automatically detected.

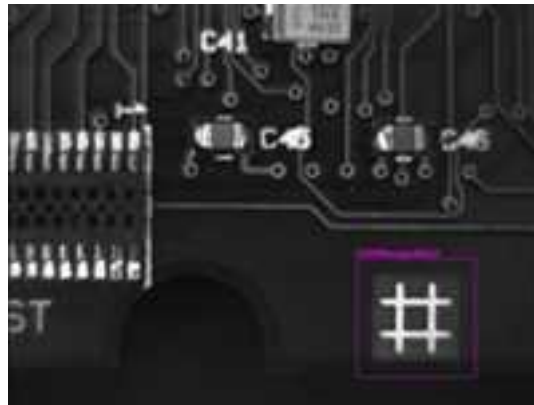
Improving the Score of Found Instances by Using "Don't Care Areas"

Setting Search Parameters

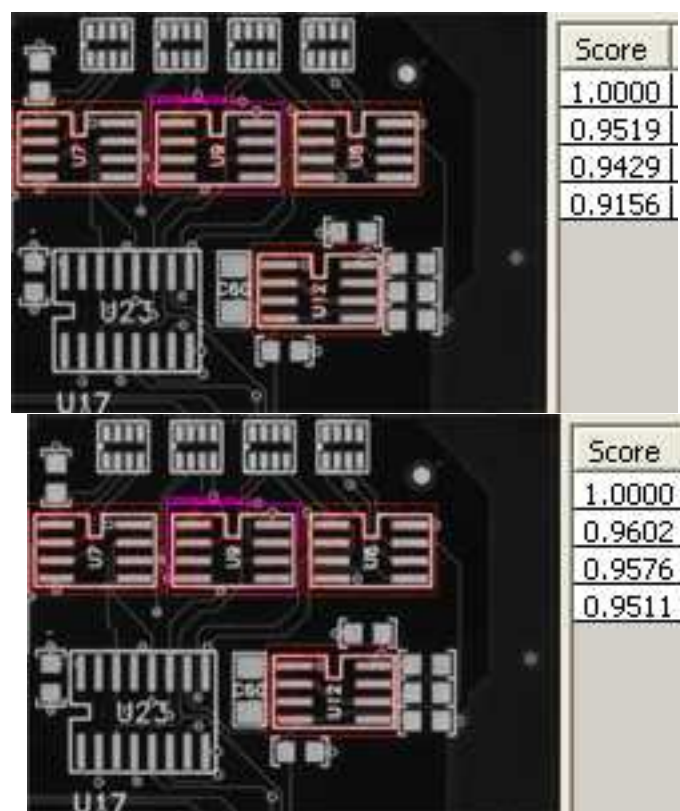
오브젝티브

Following this tutorial, you will learn how to use EasyFind to handle "don't care areas" in geometric pattern matching. "Don't care areas" help to define in the image the meaningful features only, by masking the areas that might change from image to image, such as text and numbers.

You'll need first to load a reference image, define an ROI where EasyFind will learn the reference model, and set a rotation tolerance for the expected instances to search for (steps 1-4). Then you're ready to perform automatic detection of instances of the reference model, without using "don't care areas" (step 5). As the matching scores of the found instances are not high enough, you'll define a "don't care area" on the reference model, and restart the detection. The matching scores are slightly better (steps 6-7).



Reference model



Found instances and matching scores, without (left) and with (right) using "don't care areas"

Step 1: Loading the reference image

1. From the main menu, click **EasyFind**, then **New EasyFind Tool**.
2. Keep the default variable name for the new PatternFinder object, and click **OK**. The PatternFinder management dialog box is opened.
3. In the **Model** tab, click the **Open** icon, and load the image file EasyFind\Solder Pad 1.tif.
4. Keep the default variable name for the new Image object, and click **OK**.

Step 2: Creating an ROI to define the reference model on the reference image

1. In the image, right-click and select **New ROI...** item from the menu.

2. Keep the default variable name for the new ROI object, and click **OK**. A default ROI is placed over the image (blue rectangle with handles). The ROI management dialog box is opened.
3. Drag the ROI over the reference model and resize it using its handles. Alternatively, enter the following coordinates in the ROI dialog box: 200, 130, 190, 130 for OrgX, OrgY, Width, and Height respectively, and click **Close**.

Step 3: Learning the reference model

1. In the PatternFinder **Model** tab, select the ROI object from the source image drop-down list, and click **Learn**. The reference model is detected.
2. In the PatternFinder **Search Field** tab, select the Image object from the source image drop-down list. Tick the **Draw Features** check-box. The model location and feature points are highlighted in the source image.

Step 4: Setting a rotation tolerance

1. In the PatternFinder **Allowances** tab, set the angle tolerance to 5.0.

Step 5: Detecting instances of the reference model without "don't care areas"

1. In the PatternFinder **Search Field** tab, set **Max Instances** to 4, and click **Find**. The instances matching the reference model are highlighted.
2. Click **Results** to see the matching score of each found instance. Even though the scores are good, we can still improve them slightly by using a "don't care area" to mask the text appearing in the learned pattern.

Step 6: Defining the "don't care area"

1. In the PatternFinder **Don't Care Areas** tab, select the **Rectangle** radio button from the **Blacken Inside** group.
2. In the ROI defining the reference model, move your mouse pointer on the top-left corner of the text "U9", left-click and drag the don't care area (rectangle with red stripes) to mask out the text.

Step 7: Detecting instances of the reference model with "don't care areas"

1. In the PatternFinder **Don't Care Areas** tab, click **Learn**, and then click **Find**.

The instances matching the reference model are still highlighted, but the text is not found anymore.

2. Click **Results** to compare the new matching scores.

They are slightly better.

2.7. EasyGauge

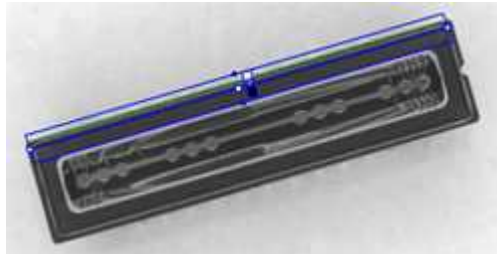
Measuring the Rotation Angle of an Object

Line Fitting

오브젝티브

Following this tutorial, you will learn how to use EasyGauge to measure the rotation angle of a CCD sensor package. As we only need to retrieve an angle value, it's not required to work in a calibrated field of view. All geometrical parameters and results will be expressed as numbers of pixels.

You'll need to load the source image (step 1), and attach a line fitting gauge (step 2). The inspection is automatically performed (step 3).



Line fitting gauge

Step 1: Load the source image

1. From the main menu, click **EasyGauge**, then **New World Shape**.
2. Keep the default variable name, and click **OK**.
3. From the **Gauges** tab, click the **Open** icon, and load the image file EasyImage\CCD.tif.
4. Keep the default variable name for the new image object, and click **OK**.

Step 2: Attach a line gauge to the image

1. In the **Gauges** tab of the world shape dialog box, right-click the world shape icon, select **New** > **Line Gauge** from the contextual menu.
2. Keep the default variable name, and click **OK**. The line location gauge appears on the image. It consists of the following elements:
 - A blue line segment along which the transitions search is carried out.
 - Five white handles, allowing the user to move and rotate the segment.
 - A gray arrow, indicating in which direction the segment is traversed.
 - Black and white rectangles, indicating which kind of transition is searched for.
 - Green line, indicating the transition points found (if any).
3. Using the handles, move, rotate and extend the line gauge, so that it is positioned on the upper edge of the CCD package, with the gray arrow pointing downwards.
4. In the **Measurement** tab of the line gauge dialog box, choose 'White to Black' from the Type dropdown list and 'From Begin' from the Choice dropdown list.

Step 3: Perform the inspection

1. The image is automatically inspected. However, clicking **Process** in the world shape dialog box will insert the corresponding code into the script window.
2. Click the **Results** tab to retrieve the measured angle value.
3. To see the individual fitting points, select the **Points** checkbox under the Draw Samples area.

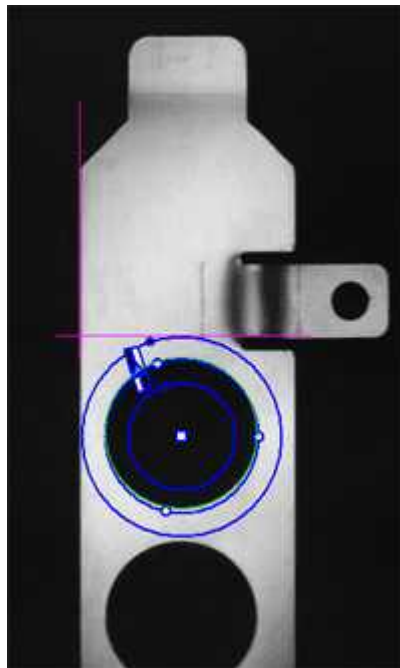
Measuring the Diameter of a Circle

Circle Fitting

오브젝티브

Following this tutorial, you will learn how to use EasyGauge to measure the diameter of a circle in an image.

You'll first load an image for calibration—a dot grid—(step 1), and calibrate the field of view (step 2). Then you'll load the source image for inspection (step 3), and attach a circle fitting gauge (step 4). The inspection is automatically performed (step 5). All measurement results are expressed in physical units..



Measuring the diameter of a circle

Step 1: Load the calibration image

1. From the main menu, click **EasyGauge**, then **New World Shape**.
2. Keep the default variable name, and click **OK**.
3. In the **Dot Grid Calibration** tab, click the **Open** icon, and load the image file EasyGauge\Dot Grid 1.tif.
4. Keep the default variable name for the new image object, and click **OK**.

Step 2: Calibrate the field of view

- Click **Calibrate**. From now on, the field of view is calibrated, and all results will be expressed in physical units.

Step 3: Load the source image

1. From the **Gauges** tab, click the **Open** icon, and load the image file EasyGauge\Bracket 6.tif.

2. Keep the default variable name, and click **OK**.

Step 4: Attach a circle gauge to the image

1. In the **Gauges** tab of the world shape dialog box, right-click the frame shape icon, select **New** > **Circle Gauge** from the contextual menu.
2. Keep the default variable name, and click **OK**.
3. The circle location gauge appears on the image. It consists of the following elements:
 - A blue ring in which the circle is searched for.
 - A blue nominal circle.
 - Six white handles, allowing the user to move and rotate the segment.
 - A gray arrow, indicating in which direction the segment is traversed.
 - Black and white rectangles, indicating which kind of transition are searched for.
 - A green arc of circle, indicating the circle found (if any).
4. Using the handles, drag the circle fitting gauge around the upper bracket hole. Adjust the nominal circle on the hole edge and extend the searching area if necessary.
5. In the **Measurement** tab of the circle gauge dialog box, select 'From Begin' from the Choice dropdown list.

Step 5: Perform the inspection

1. The image is automatically inspected. However, clicking **Process** in the world shape dialog box will insert the corresponding code into the script window.
2. Click the **Results** tab to retrieve the measured diameter. All measurement results are expressed in physical units.

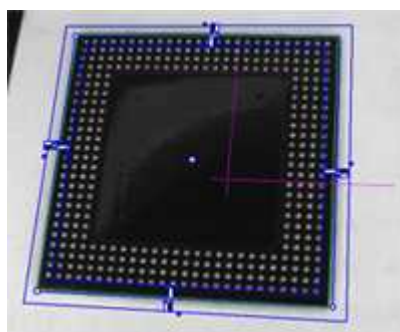
Measuring a Distorted Rectangle

Rectangle Fitting

오브젝티브

Following this tutorial, you will learn how to use EasyGauge to perform measurements on a distorted rectangle component.

To obtain measurement results in physical units, we need to work in a calibrated field of view. You'll need first to load an image for calibration—a dot grid—(step 1), and calibrate the field of view (step 2). Then you'll load the distorted image (step 3), and attach a rectangle fitting gauge (step 4). The inspection is automatically performed (step 5). All measurement results are expressed in physical units.



Measuring a distorted rectangle

Step 1: Load the calibration image

1. From the main menu, click **EasyGauge**, then **New World Shape**.
2. Keep the default variable name, and click **OK**.
3. In the **Dot Grid Calibration** tab, click the **Open** icon, and load the image file EasyGauge\Dot Grid 5.tif. This dot grid has been acquired in the same conditions and has the same distortion as the image we want to inspect.
4. Keep the default variable name for the new image object, and click **OK**.

Step 2: Calibrate the field of view

- Click **Calibrate**. From now on, the field of view is calibrated, and all results will be expressed in physical units.

Step 3: Load the distorted image

1. From the **Gauges** tab, click the **Open** icon, and load the image file EasyGauge\Distorted Component.tif.
2. Keep the default variable name, and click **OK**.

Step 4: Attach a rectangle gauge to the image

1. In the **Gauges** tab, right-click the world shape icon, and select **New** > **Rectangle Gauge** from the contextual menu.
2. Keep the default variable name, and click **OK**. The rectangle gauge dialog box is opened, and the rectangle gauge is drawn on the image. It consists of the following elements:
 - A blue rectangular ring in which the rectangle is searched for.
 - A blue nominal rectangle.
 - Eleven white handles, allowing the user to move and extend the search area.
 - Gray arrows, indicating in which direction segments are examined.
 - Black and white rectangles, indicating which kind of transition are searched for.
 - A green rectangle, indicating the rectangle found (if any).
3. Due to the perspective effect, the rectangle gauge doesn't look like a rectangle. Using the central handle, move the rectangle gauge in the image and observe the rectangle deformation. Due to the calibration, the rectangle gauge shape adapts to the field of view deformation. Extend the searching area, and adjust the nominal rectangle on the component edges.
4. In the **Measurement** tab of the rectangle gauge dialog box, select 'White To Black' from the Type dropdown list and 'From Begin' from the Choice dropdown list.

Step 5: Perform the inspection

1. The image is automatically inspected. However, clicking **Process** in the world shape dialog box will insert the corresponding code into the script window.
2. Click the **Results** tab to retrieve the measured X and Y sizes. All measurement results are expressed in physical units.

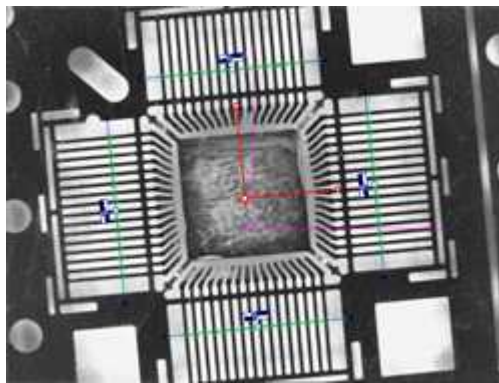
Locating Points Regarding to a Coordinate System

Point Location

오브젝티브

Following this tutorial, you will learn how to use EasyGauge to perform lead frames inspection. This operation determines the dimension, position, curvature, size, angle or diameter of the lead frames with an excellent accuracy. Robustness is ensured by powerful edge-point selection mechanisms that are intuitive and easy to tune, allowing measurement in cluttered images.

You'll first load an image for calibration —a dot grid— (step 1), and calibrate the field of view (step 2). Then you'll load the lead frame image (step 3), and set a coordinate system (a frame shape). Regarding to this coordinate system, you can define point gauges (steps 5-6). Finally, you'll load another lead frame image, that has a slight angle deviation, so the coordinate system has to be rotated (steps 7-8). The inspection is then automatically performed (step 9). All measurement results are expressed in physical units.



Four point gauges over four sets of leads

Step 1: Load the calibration image

1. From the main menu, click **EasyGauge**, then **New World Shape**.
2. Keep the default variable name, and click **OK**.
3. In the **Dot Grid Calibration** tab, click the **Open** icon, and load the image file EasyGauge\Dot Grid 2.tif.
4. Keep the default variable name for the new image object, and click **OK**.

Step 2: Calibrating the field of view

1. Click **Calibrate**. From now on, the field of view is calibrated, and all results will be expressed in physical units.

Step 3: Loading a lead frame image

1. From the **Gauges** tab, click the **Open** icon, and load the image file EasyGauge\Lead Frame 1.tif.
2. Keep the default variable name, and click **OK**.

Step 4: Setting a coordinate system

1. In the **Gauges** tab of the world shape dialog box, right-click the world shape icon, select **New** > **Frame Shape** from the contextual menu.
2. Keep the default variable name, and click **OK**. The frame shape icon appears in the world shape dialog box.
3. Drag the frame shape center approximately to the square center of the image.

Step 5: Attaching a point gauge to the frame shape

1. In the **Gauges** tab of the world shape dialog box, right-click the frame shape icon, select **New** > **Point Gauge** from the contextual menu.
2. Keep the default variable name, and click **OK**. The point location gauge appears on the image. It consists of the following elements:
 - ☐ A blue line segment along which the transitions search is carried out.
 - ☐ Three white handles, allowing the user to move and rotate the segment.
 - ☐ A gray arrow, indicating in which direction the segment is traversed.
 - ☐ Black and white rectangles, indicating which kind of transition is searched for.
 - ☐ Green crosses, indicating the transition points found (if any).
3. Place the point location gauge over a set of leads: in the **Position** tab of the point gauge dialog box, set the Center Y property to 7, and the Tolerance property to 5.

Step 6: Attaching other point gauges to the frame shape

1. From the **Gauges** tab of the world shape dialog box, create three other point gauges (refer to step 5).
2. Place these point location gauges over the remaining sets of leads :
 - ☐ Center Y = -7, Tolerance = 5
 - ☐ Center X = 7, Tolerance = 5, Angle = 90
 - ☐ Center X = -7, Tolerance = 5, Angle = 90

Step 7: Loading another lead frame image

1. From the **Gauges** tab, click the Open icon, and load the image file EasyGauge\Lead Frame 2.tif.
2. Keep the default variable name for the new image, and click **OK**.

Step 8: Tuning the coordinate system

1. In the frame shape dialog box, set the Angle property to 5.8.
2. Drag the frame shape center approximately to the square center of the image. All point location gauges automatically follow.

Step 9: Performing the inspection

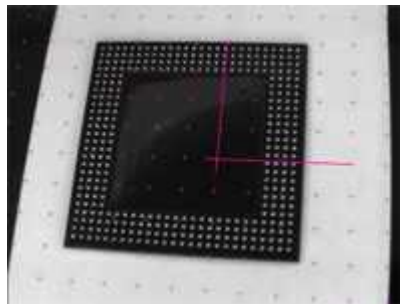
1. The image is automatically inspected. However, clicking Process in the world shape dialog box will insert the corresponding code into the script window.
2. From the point gauge dialog boxes, click the Results tab to retrieve the located points coordinates. All measurement results are expressed in physical units. The values refer to the frame shape system.

Unwarping a Distorted Image

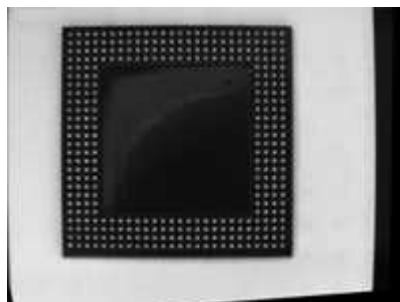
오브젝티브

Following this tutorial, you will learn how to use EasyGauge to perform grid calibration, and unwarp a distorted image.

You'll first load an image for calibration—a dot grid— (step 1), and calibrate the field of view (step 2). Then you'll load the distorted image (step 3), and perform the unwarping operation (step 4).



Distorted image



Unwarped image

Step 1: Load the calibration image

1. From the main menu, click **EasyGauge**, then **New World Shape**.
2. Keep the default variable name, and click **OK**.
3. In the **Dot Grid Calibration** tab, click the **Open** icon, and load the image file EasyGauge\Dot Grid 5.tif. This dot grid has been acquired in the same conditions and has the same distortion as the image we want to unwrap.
4. Keep the default variable name for the new image object, and click **OK**.

Step 2: Calibrate the field of view

- Click **Calibrate**.

Step 3: Load the distorted image

1. From the **Unwarping** tab, click the **Open** icon, and load the image file EasyGauge\Distorted component.tif.
2. Keep the default variable name, and click **OK**.

Step 4: Unwarp the distorted image

1. In the Destination Image area, click **New** icon to create a new image.
2. Keep the default image settings, and click **OK**.
3. Select **Interpolate** checkbox to improve resulting image quality.
4. Click **Unwarp**. In the destination image, all distortions are corrected.

2.8. EasyOCR

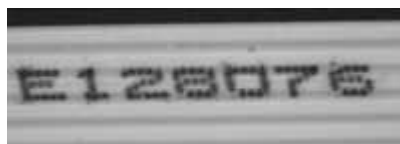
Learning Characters and Creating an EasyOCR Font

Learning Characters

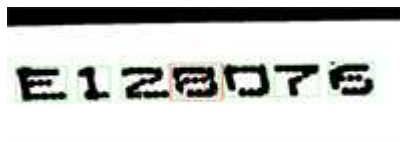
오브젝티브

Following this tutorial, you will learn how to use EasyOCR to learn new characters and save them in an EasyOCR font.

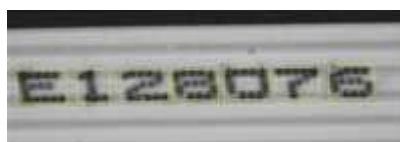
You'll need first to load a source image (step 1). Then you'll set the segmentation parameters to isolate each character (step 2). Each character will have to be learnt (step 3), and finally you'll save all the learnt characters as a font file (step 4). You can also add new characters to an existing font if needed (step 5).



소스 이미지



The image is segmented so that all the characters are detected



All the characters have been learn

Step 1: Load the source image

1. From the main menu, click **EasyOCR**, then **New OCR Tool**.
2. Keep the default variable name for the new OCR object, and click **OK**.
3. In the **Source Image** tab, click the **Open** icon of the Source Image area, and load the image file EasyOCR\FlatCable\FlatCable1.tif.
4. Keep the default variable name for the new image object, and click **OK**.

Step 2: Set segmentation parameters

1. Select the **Segmentation Parameters** tab, and move the red frame in the image above a character.
2. Tune each property to get a green bounding box around each character:
 - ☐ **threshold value** = 113
 - ☐ **characters** color = Black on White
 - ☐ **min width** = 36
 - ☐ **min height** = 31
 - ☐ **spacing** = 4
 - ☐ **max width** = 98
 - ☐ **max height** = 72
 - ☐ **noise area** = 9

Step 3: Learn new characters

1. Select the **Learn** tab, and click the character E in the image. You are then prompted to identify the character along with its class. Enter E in the character field, and select the 'E0crClass_Uppercase' class. **Ok**를 클릭하십시오. Whenever a character has been added to the current font, its bounding box turns yellow.
2. Click the character 1 in the image. Enter 1 in the character field, and select the 'E0crClass_Digit' class. **Ok**를 클릭하십시오.
3. Proceed with remaining characters.

Step 4: Save the EasyOCR font

- In the **Font File** tab, click the **Save As...** button. Type a file name for the new EasyOCR font file. Its extension will be .ocr. Finally, click **Save**.

Step 5: Add characters to an existing font

1. In the Source Image tab, click the **Open** icon of the Source Image area, and load the image file EasyOCR\FlatCable\FlatCable2.tif.
2. Keep the default variable name for the new image object, and click **OK**.
3. In the **Recognition** tab, click **Execute**. Characters 2 and 8 are read correctly, but A, W and G are not (**low confidence score**). They don't belong to the font.
4. Select the **Learn** tab, and learn the characters A, W, and G (refer to step 3).
5. Then save the font again (refer to step 4). The new characters have been added.

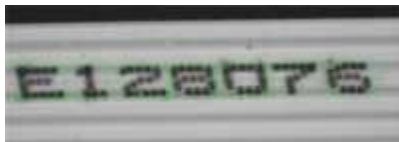
Recognizing Characters According to a Font

"Recognizing Characters According to a Font"페이지55

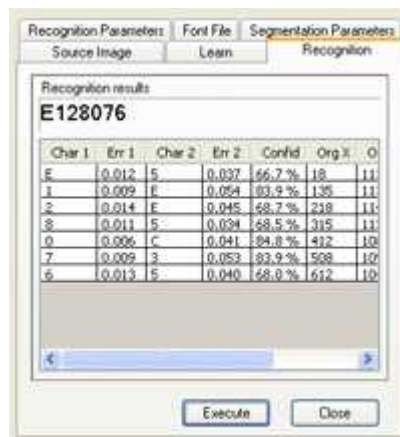
오브젝티브

Following this tutorial, you will learn how to use EasyOCR to recognize characters, regarding to a specific font.

You'll need first to load a source image (step 1), and an EasyOCR font file (step 2). Then you'll perform the characters recognition (step 3).



Characters matching the font are automatically detected



Results after explicit recognition

Step 1: Load the source image

1. From the main menu, click **EasyOCR**, then **New OCR Tool**.
2. Keep the default variable name for the new OCR object, and click **OK**.
3. In the Source Image tab, click the **Open** icon of the Source Image area, and load the image file EasyOCR\FlatCable\FlatCable1.tif.
4. Keep the default variable name for the new image object, and click **OK**.

Step 2: Load the font file

- In the **Font File** tab, click **Load**, and select the font file EasyOCR\FlatCable\FlatCable.ocr. In the image, the detected characters are highlighted in green.

Step 3: Recognize the characters

- In the **Recognition** tab, click **Execute** to trigger the recognition of the detected characters. The recognized characters appear in the Recognition results area. Further information about each character can be found in the table.

2.9. EasyBarCode

Reading Bar Codes Automatically

Reading a Bar Code

오브젝티브

Following this tutorial, you will learn how to perform automatic reading of multiple bar codes.

You'll need first to load multiple source images (step 1). The reading is then automatically performed on each image (step 2).



Each bar code is automatically detected and decoded

Step 1: Load the source images

- From the main menu, click **EasyBarCode**, then **New BarCode Tool**.
- Keep the default variable name, and click **OK**.

3. In the **AutoRead** tab, click the **Open** icon of the Source Image area, and load the image files EasyBarcode\Barcode 01.tif to Barcode 10.tif. Use the shift key to select multiple files.
4. Keep the default variable name, and click **OK**. The last image appears.

Step 2: Read the bar codes automatically

1. The bar code is automatically detected and decoded. The graphic result appears on the image, while the data content and the corresponding symbology are displayed in the Decoded Symbology area. It is not necessary to click **Read** once a new image appears. However, clicking **Read** will insert the corresponding code into the script window.
2. In the **Results** tab, find more information about the bar code. As a bar code might have a meaning under different symbologies, all possible contents are listed by decreasing likeliness.
3. In the **AutoRead** tab, click the **Load Next** and **Load Previous** icons to browse through images 01 to 06.

The image files appear, and each bar code is automatically detected and decoded. The bar code properties are updated.

4. To decode the remaining bar codes, we have to enable the additional symbologies.
5. In the **Symbologies** tab, click the **Toggle All** button of the Additional area.
6. In the **AutoRead** tab, click the **Load Next** and **Load Previous** icons and browse the remaining images.

Note about EasyBarcode2

Since the version 22.04, **Open eVision Studio** showcases **EasyBarcode2**, our new bar code reader: it is faster, reads more complex bar codes, can read several bar codes on an image and supports more symbologies.

Don't hesitate to check it out!

- To read Barcode 02 (MSI), Barcode 05 (Code 25 Interleaved), Barcode 07 (Code 25 Datalogic), Barcode 08 (Code 11), Barcode 10 (Code CIP) and Barcode 12 (PharmaCode), enable the corresponding symbology in the symbologies tab.
- To read Barcode 03 (Code 39), disable the bar code validation in the **AutoRead** tab.

2.10. EasyMatrixCode

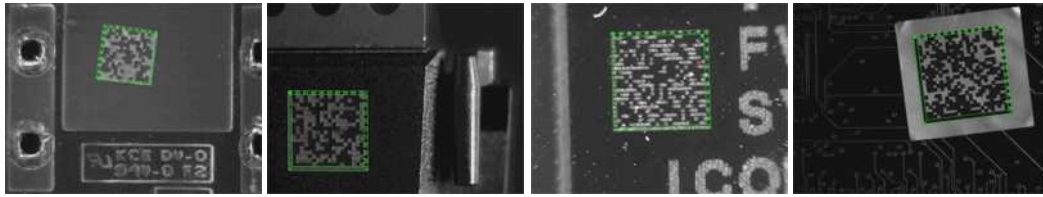
Reading Data Matrix Codes Automatically

Automatic Reading

오브젝티브

Following this tutorial, you will learn how to use EasyMatrixCode to detect and decode automatically Data Matrix codes in multiple files.

You'll need first to load multiple source images (step 1). The reading is then automatically performed on each image (step 2). You can also grade the printing quality of each matrix code (step 3).



Each Data Matrix code is automatically detected and decoded

Step 1: Load the source images

1. From the main menu, click **EasyMatrixCode**, then **New MatrixCode Tool**.
2. Keep the default variable name for the new matrix code reader, and click **OK**.
3. In the **Read** tab, click the **Open** icon of the Source Image area, and load the image files EasyMatrixCode\AutoRead\AutoRead 01.tif to AutoRead 04.tif. Use the shift key to select multiple files.
4. Keep the default variable name for the new image, and click OK. The last image appears.

Step 2: Read the Data Matrix codes automatically

1. The Data Matrix code is automatically detected and decoded. The matrix code reference corner is highlighted with a bold cross mark. It is not necessary to click **Read** once a new image appears. However, clicking **Read** will insert the corresponding code into the script window.
2. In the **Results** area, find more information about the matrix code, such as the decoded string.
3. In the **Read** tab, click the **Load Next** and **Load Previous** icons. The image files appear, and each Data Matrix code is automatically detected and decoded. The matrix code properties are updated. If no Data Matrix code could be located in the image, an error message is displayed in the Message field.

Step 3: Grade Data Matrix code printing quality

1. In the **Print Quality** tab, select the **Compute Grading** check-box.
2. Click **Apply**.

For each printing quality parameter, the corresponding value and its grade equivalent appear. An A grade means a good quality, while an F grade indicates a poor one.

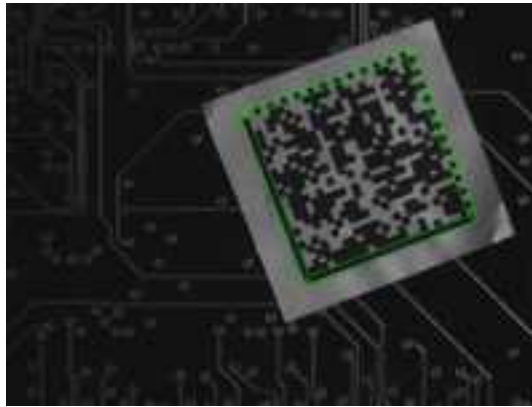
Learning a Data Matrix Code and Creating an EasyMatrixCode Model File

Reading with Prior Learning

오브젝티브

Following this tutorial, you will learn how to use EasyMatrixCode to learn a Data Matrix code, and save it as an EasyMatrixCode model file.

You'll need first to load a source image (step 1), and learn the matrix code (step 2). Then you'll save the learned matrix code as a new model file (step 3). You can also add new learned matrix codes to an existing model if needed (step 4).



The Data Matrix code has been learned

Step 1: Load the source image

1. From the main menu, click **EasyMatrixCode**, then **New MatrixCode Tool**.
2. Keep the default variable name for the new matrix code reader object, and click **OK**.
3. In the **Learn** tab, click the **Open** icon of the Source Image area, and load the image file `EasyMatrixCode\Label\Label 4.tif`.
4. Keep the default variable name for the new image object, and click **OK**.

Step 2: Learn the Data Matrix code

- In the **Learn** tab, click **Learn**.

The Data Matrix code is detected and decoded without error.

The graphical result appears on the image.

The properties of the learned matrix code are updated in the dialog box.

Step 3: Save the model file

1. In the **Learn** tab, click the **Save As...** button.
2. Type a file name for the new EasyMatrixCode model file. Its extension will be `.mx2`.
3. **저장**을 클릭합니다.

Step 4: Learning more Data Matrix codes

1. In the **Read** tab, click the **Open** icon of the source image area, and load the image file `EasyMatrixCode\PCB Code\PCB Code 3.jpg`.
2. Keep the default variable name for the new image object, and click **OK**.

3. An error message is displayed in the Message area of the [Read](#) tab. The matrix code can not be read, since the reader uses the model from the "Label 4" image. You need to learn the "PCB Code 3" matrix code, and add it to the model.
4. In the [Learn](#) tab, click [Learn More](#). The Data Matrix code is detected and decoded without error. The graphical result appears on the image. The properties of the learned matrix code are updated in the dialog box.
5. Using [Learn More](#) rather than Learn involves that the "Label 4" model is not replaced by the "PCB Code 3" model, but both are now included in the same model. In the Read tab, the "PCB Code 3" matrix code is correctly read. Select the "Label 4" image in the drop-down list of the source image area. The "Label 4" matrix code is still read without error, which means that both learned matrix codes have been kept.
6. Finally, save the model again (refer to step 3). The new matrix code has been added.